Guided Lecture Notes

for

Proof, Set Theory, and Logic

MATH 330

Complete packet

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These guided notes accompany the Fall 2026 revised course edition of *Elementary Set Theory: Proof, Set Theory, and Logic*. They are based on Richard P. Millspaugh's original textbook materials and lecture materials prepared and revised by Bryce A. Christopherson.

The blank spaces in the student copy are completed during class. Each blank occupies exactly the same space as the corresponding typed completion in the instructor copy, so the two packets have identical pagination. Definitions, theorem statements, and the surrounding explanations are provided so that class time can focus on examples, proof decisions, and the mathematical steps that connect them.

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Chapter 1

Elementary Logic

1.1 Propositions

Definition 1.1.
A *proposition* is a statement that is either true or false, but not both. The *truth value* of a proposition is true (T) if the sentence is true and false (F) if the sentence is false.

Example 1.2.
Determine whether the following are propositions:

- *Two plus two equals four.*

  This sentence is true, hence a proposition.

- *Seven minus three equals twelve.*

  This sentence is false, hence a proposition.

- *It will snow in Grand Forks on March 17, 2525.*

  This sentence is a proposition; we just don't happen to know the truth value.

- *Go clean your room.*

  This sentence is a command, not a proposition.

- *Is it raining outside?*

  This sentence is not a proposition. It is a question.

- *This sentence is false.*

  This sentence is not a proposition because it cannot be either true or false. If it is true, then its claim is false. If it is false, then its claim must be true.

- $x + 3 = 12$.

  This sentence cannot be said to be true or false without more information. If $x = 0$, then it is false. If $x = 9$, then it is true. Statements like this are not propositions.

We will use uppercase letters, frequently P or Q, to stand for variable propositions in much the same way that you might use x or y to represent variable numbers in algebra.

1.2 Connectives and Truth Tables

Many of the propositions we are interested in are made up of more elementary propositions. For example, the proposition S: *today is Tuesday and it's raining* is made up of the two propositions P: *today is Tuesday* and Q: *it's raining*. The word *and* in S is called a connective since it gives us a way to connect two propositions and form another. In this section we will look at several kinds of connectives.

1.2.1 Conjunction

Definition 1.3.
The *conjunction* of P and Q, denoted $P \wedge Q$, is true exactly when both P and Q are true.

We will sometimes keep track of truth values in a *truth table*, where we list the truth value of a proposition in terms of the truth values of its elementary propositions.

Example 1.4.
Construct a truth table for the conjunction $P \wedge Q$.

P	Q	$P \wedge Q$
T	T	T
T	F	F
F	T	F
F	F	F

Note that in this case there are four rows in the table. For a proposition involving n distinct elementary propositions, there are 2^n possible truth assignments and hence 2^n rows in its truth table. This makes it cumbersome to construct truth tables for very complicated propositions. Nevertheless, we will find them to be a convenient tool.

1.2.2 Disjunction

We next consider propositions of the form P *or* Q. The word *or* can be ambiguous in English: sometimes it includes the possibility that both conditions hold, while in other contexts only one option is intended. In logic we use *or* inclusively. Thus $P \vee Q$ is true when at least one of P and Q is true.

Definition 1.5.
The *disjunction* of P and Q, denoted $P \vee Q$, is true whenever at least one of P or Q is true.

Example 1.6.

Construct a truth table for the proposition $P \vee Q$.

P	Q	$P \vee Q$
T	T	T
T	F	T
F	T	T
F	F	F

1.2.3 Implication

Most mathematical propositions have the form *If* P, *then* Q. Of course, either of P or Q might be propositions made of several other propositions.

Definition 1.7.
The proposition *If* P, *then* Q is called an *implication* and is denoted $P \Rightarrow Q$. It is false exactly when P is true and Q is false.

Example 1.8.

Construct a truth table for $P \Rightarrow Q$.

P	Q	$P \Rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

Notice: To say that $P \Rightarrow Q$ is true **does not** mean that P is true or that Q is true. It **does** mean that there is a relationship between the truth values of these two propositions. The implication $P \Rightarrow Q$ is false when P is true and Q is false; it is true when either P is false or Q is true.

There are several other implications related to the implication $P \Rightarrow Q$ that we will occasionally find useful.

Definition 1.9.
For the implication $P \Rightarrow Q$:

- The *converse* is $Q \Rightarrow P$.
- The *contrapositive* is $\neg Q \Rightarrow \neg P$.
- The *inverse* is $\neg P \Rightarrow \neg Q$.

1.2.4 Logical Equivalence

Definition 1.10.
The *biconditional* $P \Leftrightarrow Q$, read "P if and only if Q," is true exactly when P and Q have the same truth value under the given assignment. Two propositional forms P and Q are *logically equivalent* if they have the same truth value under every assignment of truth values to their elementary propositions. Equivalently, $P \Leftrightarrow Q$ is a tautology.

Example 1.11.
Construct a truth table for $P \Leftrightarrow Q$.

P	Q	$P \Leftrightarrow Q$
T	T	T
T	F	F
F	T	F
F	F	T

1.2.5 Negation

Definition 1.12.
The *negation* of P, denoted $\neg P$ and sometimes read "not P," is the proposition whose truth value is always opposite that of P.

Negation is a unary connective: it acts on one proposition, whereas conjunction, disjunction, implication, and biconditional are binary connectives.

Example 1.13.
Construct a truth table for P and $\neg P$.

P	$\neg P$
T	F
F	T

1.2.6 Tautology and Contradiction

Definition 1.14.
A *tautology* is a proposition that is always true, regardless of the truth values of the elementary propositions that make it up.

Definition 1.15.
A *contradiction* is a proposition that is always false.

EXAMPLE 1.16.

Show the proposition $P \vee \neg P$ is a tautology and that the proposition $P \wedge \neg P$ is a contradiction using a truth table.

P	$\neg P$	$P \vee \neg P$	$P \wedge \neg P$
T	F	T	F
F	T	T	F

EXAMPLE 1.17.

Use a truth table to show that $P \Rightarrow (P \vee Q)$ is a tautology.

We construct a column for $P \Rightarrow (P \vee Q)$ and check that every entry in that column is T.

P	Q	$P \vee Q$	$P \Rightarrow (P \vee Q)$
T	T	T	T
T	F	T	T
F	T	T	T
F	F	F	T

EXAMPLE 1.18.

Use a truth table to show that $(P \Rightarrow Q) \Leftrightarrow (\neg Q \Rightarrow \neg P)$ is a tautology.

It is enough to compare the columns for $P \Rightarrow Q$ and $\neg Q \Rightarrow \neg P$ and verify that they agree in every row.

P	Q	$\neg P$	$\neg Q$	$P \Rightarrow Q$	$\neg Q \Rightarrow \neg P$
T	T	F	F	T	T
T	F	F	T	F	F
F	T	T	F	T	T
F	F	T	T	T	T

1.3 Predicates, Instantiation, and Quantification

We return our attention now to statements that involve variables, like $x + 3 = 12$. Such a statement is called a *predicate* and we will usually indicate it with $P(x)$ rather than P. Before a predicate takes on a truth value, we need to know something about the variable. We noted before that if $x = 0$, then the statement is false; but if $x = 9$, then the statement is true. What if x represents my desk? In that case the statement is complete nonsense, but you probably feel like there's something a little bit unfair about that choice for x. We see an equation and usually think that the variable must represent a number, right? The first thing you should know about any variable in a predicate is what it can represent. This is called the *universe* or the *domain of discourse*. In mathematics, we will be very explicit about what universe we are talking about.

1.3.1 Instantiation

Suppose the universe for the preceding predicate $P(x)$ is $\mathbb{R}$. That still doesn't turn $P(x)$ into a proposition because, as noted above, $P(x)$ doesn't take on a truth value until we know which real number x indicates. One way to do this is called *instantiation*, which means choosing a particular value for the variable(s). Consider for example the sentence: *Let* $x = 2$, *then* $x + 3 = 12$. In this case the sentence is a proposition with truth value F. The other option is *quantification* of the variable(s). We will discuss two kinds of quantifiers, *universal* and *existential*. You might occasionally encounter other quantifiers, but they are not necessary.

1.3.2 Universal Quantification

Definition 1.19.
A *universal quantifier* indicates that the predicate is true for all instances of the variable in the given universe. It is typically indicated by using the phrase "for every" or "for all," and can be denoted symbolically by $\forall$ in symbolic statements.

The following sentences are universally quantified propositions. Only the second is true.

- *For every real number* x, $x + 3 = 12$.
- *The square of* x *is nonnegative for all real* x.
- *The square of* x *is real for all complex* x.

The third statement is false; for example, $(1 + i)^2 = 2i$, which is not real.

1.3.3 Existential Quantification

Definition 1.20.
An *existential quantifier* indicates that the predicate is true for at least one instance of the variable in the given universe. It is typically indicated by the phrase "for some" or "there exists," and can be denoted by $\exists$ in symbolic statements.

The following sentences are existentially quantified. All three of these are true.

- *There is a real number x so that $x + 3 = 12$.*
- *The square of x is nonnegative for some real x.*
- *There is a complex number x whose square is -1.*

1.3.4 Syllogisms and Venn Diagrams

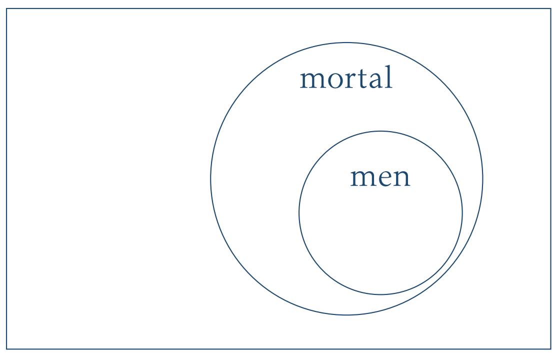
A *syllogism* is a form of logical argument that deduces a conclusion based on two premises. For example:

All men are mortal. Socrates is a man. Hence Socrates is mortal.

This is a valid syllogism. If we assume that the premises are true, then the conclusion must follow. One tool we can use to help think about syllogisms is a *Venn diagram*, which in its simplest form is a collection of circles that represent the categories in the premises.

Example 1.21.
Construct a Venn diagram to determine if the following syllogism is valid: *All men are mortal. Socrates is a man. Hence Socrates is mortal.*

In this case we will have one circle representing mortals and another representing men. Since one of our premises is that all men are mortal, the circle representing men is completely contained inside the circle representing mortals.



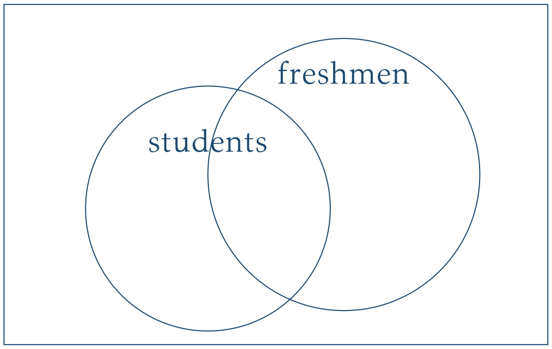
Since Socrates is a man, he lies inside the circle representing men. That circle is contained in the circle representing mortals, so Socrates is mortal and the syllogism is valid.

Example 1.22.

Determine whether the following syllogism is valid:

Some students are freshmen. Pat is a student. Hence Pat is a freshman.

The circle for students should overlap the circle for freshmen, but need not be contained in it, since the premise says only that some students are freshmen. Here is a Venn diagram:



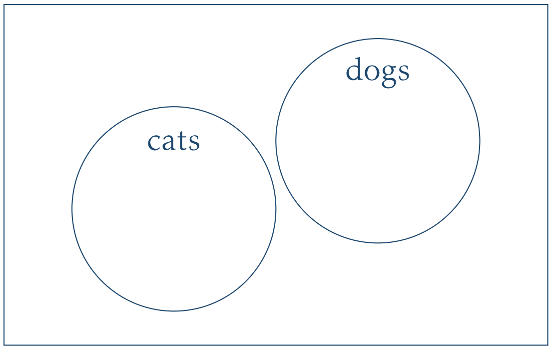
Pat may lie in the student circle without lying in the freshman circle, so the syllogism is not valid.

Example 1.23.

Determine whether the following syllogism is valid:

No cats are dogs. Jade is a cat. Hence Jade is not a dog.

In this case, the circle for cats should be completely outside the circle for dogs since no cats are dogs. Here is a Venn diagram:



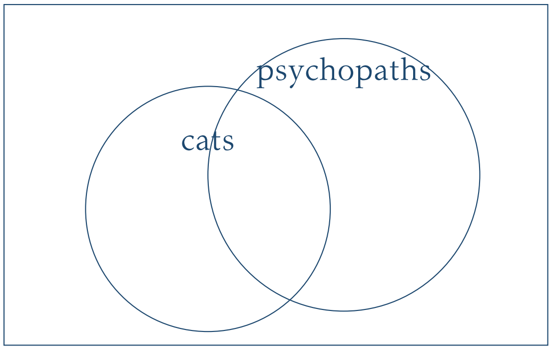
Since Jade lies in the cats circle, Jade cannot lie in the dogs circle, so the syllogism is valid.

Example 1.24.

Determine whether the following syllogism is valid:

Some cats are psychopaths. Jawa is a cat. Hence Jawa is a psychopath.

Here is a Venn diagram:



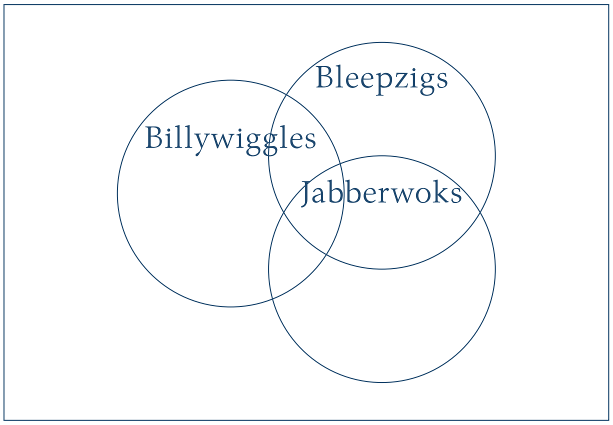
Jawa lies in the cats circle but may or may not lie in the psychopaths circle, so this syllogism is not valid.

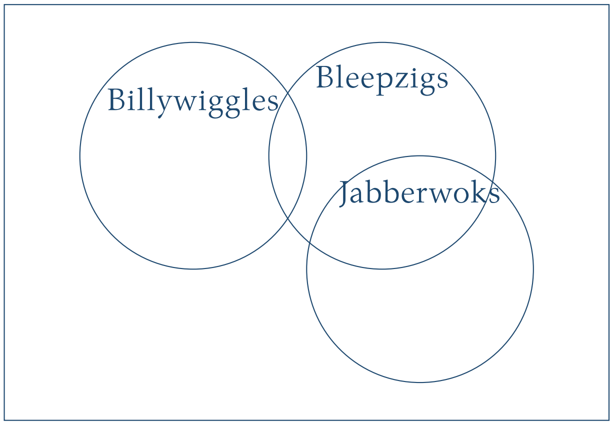
Example 1.25.

Determine whether the following syllogism is valid:

Some Billywiggles are Bleepzigs. Some Bleepzigs are Jabberwoks. Hence some Billywiggles are Jabberwoks.

It is possible for the Venn diagram to look like either of the following:





This syllogism is not valid.

1.3.5 Quantification of Several Variables

A predicate can have two or more variables. In order to use the predicate in a proposition, a universe must be declared for each variable and each variable must be instantiated or quantified. We will discuss the various ways a two-variable predicate might be quantified by considering an example.

Example 1.26.

Consider the predicate $P(x, y)$ given by $x + y = 0$ and assume that the universe for both x and y is the set of real numbers. Determine whether the following proposition is true: $\forall x \, \forall y \, P(x, y)$

This proposition can be read as *for all (real) x and for all (real) y*, $x + y = 0$, which says the sum of any two real numbers is 0. Clearly this is false.

Example 1.27.

Consider the predicate $P(x, y)$ given by $x + y = 0$ and assume that the universe for both x and y is the set of real numbers. Determine whether the following proposition is true: $\forall y \, \forall x \, P(x, y)$.

Restating this as we did above, it is not difficult to see that this proposition is equivalent to the first one.

Example 1.28.

Consider the predicate $P(x, y)$ given by $x + y = 0$ and assume that the universe for both x and y is the set of real numbers. Determine whether the following proposition is true: $\exists x \, \exists y \, P(x, y)$.

We can read this as *there is an x so that there is a y so that* $x + y = 0$, which says that there are real numbers x and y such that $x + y = 0$. This is true. Once again, reversing the order of the quantifiers doesn't matter.

Example 1.29.

Consider the predicate $P(x, y)$ given by $x + y = 0$ and assume that the universe for both x and y is the set of real numbers. Determine whether the following proposition is true: $\forall x\, \exists y\, P(x, y)$.

This proposition says that *for every x there is a y such that $x + y = 0$*, in other words every real number has an additive inverse, which is certainly true.

Example 1.30.

Consider the predicate $P(x, y)$ given by $x + y = 0$ and assume that the universe for both x and y is the set of real numbers. Determine whether the following proposition is true: $\exists y\, \forall x\, P(x, y)$.

The variables here are quantified the same, but the order is changed. Does that matter? The proposition this time says *there is a y so that for all x, $x + y = 0$*. In other words, there is some real number y so that adding any real number to y yields a sum of 0, which is certainly false!

1.4 Negations

The negation of a proposition is the proposition with the opposite truth value. In this section we are going to discuss the negations of compound propositions, which can be an important step in some kinds of proofs. We begin with two theorems that tell how to find the negations of propositions involving connectives.

Theorem 1.31.
Given any propositions P *and* Q, $\neg(P \Rightarrow Q) \Leftrightarrow (P \wedge \neg Q)$.

Proof. Complete the truth table below and compare its final two columns.

P	Q	$\neg Q$	$P \Rightarrow Q$	$\neg(P \Rightarrow Q)$	$P \wedge \neg Q$
T	T	F	T	F	F
T	F	T	F	T	T
F	T	F	T	F	F
F	F	T	T	F	F

The final two columns agree in every row, so $\neg(P \Rightarrow Q)$ and $P \wedge \neg Q$ are logically equivalent.

Theorem 1.32 (De Morgan's Laws)**.**
Given any propositions P *and* Q*:*

(i) $\neg(P \wedge Q) \Leftrightarrow (\neg P \vee \neg Q)$

(ii) $\neg(P \vee Q) \Leftrightarrow (\neg P \wedge \neg Q)$

You will prove this theorem in the homework.

Example 1.33.
Negate the proposition $P \Rightarrow (Q \wedge R)$.

We apply what we know one step at a time, beginning with the implication:

$$\begin{aligned}\neg\big(P \Rightarrow (Q \wedge R)\big) &\Leftrightarrow P \wedge \big(\neg(Q \wedge R)\big) && \text{(Theorem 1.31)}\\ &\Leftrightarrow P \wedge (\neg Q \vee \neg R) && \text{(Theorem 1.32(i))}\end{aligned}$$

Example 1.34.
Find the negation of $P \wedge (Q \Rightarrow R)$.

We apply Theorem 1.31 and Theorem 1.32 where indicated.

$$\begin{aligned}\neg\big(P \wedge (Q \Rightarrow R)\big) &\Leftrightarrow \big(\neg P \vee \neg(Q \Rightarrow R)\big) && \text{(Theorem 1.32(i))}\\ &\Leftrightarrow \big(\neg P \vee (Q \wedge \neg R)\big) && \text{(Theorem 1.31)}\end{aligned}$$

1.4.1 Negating Quantified Statements

The quantifiers $\forall$ and $\exists$ are interchanged by negation. This is the quantified analogue of De Morgan's Laws.

Theorem 1.35 (Quantified Negation). *Let* $P(x)$ *be a predicate on a specified domain. Then*

$$\neg\big(\forall x\, P(x)\big) \iff \exists x\, \neg P(x)$$

and

$$\neg\big(\exists x\, P(x)\big) \iff \forall x\, \neg P(x).$$

Proof.

The statement $\forall x\, P(x)$ is false exactly when at least one object in the domain fails to satisfy P; this is precisely $\exists x\, \neg P(x)$. Likewise, $\exists x\, P(x)$ is false exactly when no object satisfies P, meaning that every object satisfies $\neg P$.

For nested quantifiers, move the negation inward one quantifier at a time. For example,

$$\neg\big(\forall x\, \exists y\, Q(x,y)\big) \iff \exists x\, \forall y\, \neg Q(x,y).$$

Notice that in $\forall x\, \exists y\, Q(x,y)$ the witness y may depend on x; reversing the quantifiers would assert the existence of one choice of y that works for every x.

Example 1.36.

Find the negation of $(P \vee Q) \wedge (P \vee R)$.

We apply Theorem 1.32 where indicated.

$$\begin{aligned}\neg\big((P \vee Q) \wedge (P \vee R)\big) &\Leftrightarrow \big(\neg(P \vee Q) \vee \neg(P \vee R)\big) && \text{(Theorem 1.32(i))}\\ &\Leftrightarrow \big((\neg P \wedge \neg Q) \vee (\neg P \wedge \neg R)\big) && \text{(Theorem 1.32(i,ii))}\end{aligned}$$

Chapter 2

Logical Arguments

2.1 Rules of Inference

We accept the following two basic rules of thought:

> The *Law of Excluded Middle* says that given any proposition P, we may deduce $P \vee \neg P$. In words, either P is true or $\neg P$ is true.
>
> The *Law of non-Contradiction* says that given any proposition P, we may deduce $\neg(P \wedge \neg P)$. In words, P and $\neg P$ cannot both be true.

A *rule of inference* is a logical rule consisting of a hypothesis and a conclusion. Whenever the hypothesis is true, the conclusion must also be true. For our purposes this means that once we have deduced all of the hypotheses, we may also deduce the conclusion. Rules of inference are closely related to tautologies. In fact, they are essentially the same thing. While every tautology gives rise to a rule of inference, in practice the seven listed in the table below will usually suffice.

Table 2.1: Rules of Inference

Name	**Hypotheses**	**Conclusion**
Modus ponens	P and $P \Rightarrow Q$	Q
Modus tollens	$\neg Q$ and $P \Rightarrow Q$	$\neg P$
Hypothetical syllogism	$P \Rightarrow Q$ and $Q \Rightarrow R$	$P \Rightarrow R$
Addition	P	$P \vee Q$
Simplification	$P \wedge Q$	P
Conjunction	P and Q	$P \wedge Q$
Disjunctive syllogism	$P \vee Q$ and $\neg P$	Q

The Rule of Addition, for example, comes from the tautology $P \Rightarrow (P \vee Q)$. Since the implication is always true, knowing that the hypothesis is true will tell us that the conclusion must also be true. We can verify each of the Rules of Inference in Table 2.1 by constructing a truth table for the associated tautology.

2.1.1 Propositional Arguments

Each of our Rules of Inference is actually a *propositional argument*, where accepting the hypotheses forces us to accept the conclusion. We use the following notation for propositional arguments, where the propositions above the line are hypotheses and the proposition(s) below the line are the conclusion(s). We precede the conclusion(s) with the symbol $\therefore$, which is read "therefore."

Example 2.1.

Here is the rule of modus tollens as a propositional argument:

$$\begin{array}{c} \neg Q \\ P \Rightarrow Q \\ \hline \therefore \neg P \end{array}$$

Example 2.2.

Express the rule of modus ponens as a propositional argument.

$$\begin{array}{c} P \\ P \Rightarrow Q \\ \hline \therefore Q \end{array}$$

Example 2.3.

Express the rule of hypothetical syllogism as a propositional argument.

$$\begin{array}{c} P \Rightarrow Q \\ Q \Rightarrow R \\ \hline \therefore P \Rightarrow R \end{array}$$

Definition 2.4.
A propositional argument is said to be *valid* if accepting the hypotheses forces us to accept the conclusion(s).

How do we determine whether or not a propositional argument is valid? One option is to construct a truth table, but that can be very tedious. Our preferred option is to construct a logical argument which begins by assuming each of the hypotheses is true and deduces consequences using our rules of inference. Each deduced proposition is a step in our proof that can be justified as either a hypothesis, a tautology, a consequence of previous steps and a rule of inference, or a proposition that is logically equivalent to a previous step. The proof is complete when we have deduced the desired conclusion.

Example 2.5.
Prove that the following argument is valid:

$$\begin{array}{c} P \\ P \Rightarrow Q \\ (Q \vee R) \Rightarrow S \\ \hline \therefore S \end{array}$$

Proof.

(1)	P	hypothesis
(2)	$P \Rightarrow Q$	hypothesis
(3)	Q	modus ponens, (1) and (2)
(4)	$Q \vee R$	addition, (3)
(5)	$(Q \vee R) \Rightarrow S$	hypothesis
(6)	S	modus ponens, (4) and (5)

Example 2.6.

Prove that the following argument is valid:

$$\frac{P \Rightarrow (Q \wedge \neg Q)}{\therefore \neg P}$$

Proof.

(1)	$P \Rightarrow (Q \wedge \neg Q)$	hypothesis
(2)	$\neg(Q \wedge \neg Q)$	non-Contradiction
(3)	$\neg P$	modus tollens, (1) and (2)

The argument in Example 2.6 says that if a proposition P implies a contradiction, then $\neg$P must be true. This result is the basis for proofs by contradiction.

2.1.2 Arguments with Quantifiers

Arguments involving quantified propositions require rules for instantiation and generalization. The *rules for instantiation* let us deduce statements about particular elements of the domain from quantified statements. The *rules for generalization* tell us when deductions about elements of the domain justify quantified statements. We give these rules in Table 2.2.

Table 2.2: Rules of Quantification

Name	**Rule**
Universal Instantiation	$\forall x\ P(x) \therefore P(c)$ whenever c is in the domain of x
Existential Instantiation	$\exists x\ P(x) \therefore P(c)$ for some particular but unspecified c in the domain of x
Universal Generalization	$P(c)$ for *arbitrary* c in the domain of $x \therefore \forall x\ P(x)$
Existential Generalization	$P(c)$ for *some* c in the domain of $x \therefore \exists x\ P(x)$

The restrictions in this table matter. In existential instantiation, the letter c denotes one particular but unspecified value whose existence is guaranteed by the existential statement. We do not get to choose which value it is, and we may not assume anything about it beyond what the existential statement tells us. The letter c should therefore not already be in use elsewhere in the argument. In universal generalization, by contrast, c represents an arbitrary element of the domain.

Example 2.7.

Prove that the following argument is valid:

$$\begin{array}{c} \forall x\ (\mathrm{P}(x) \wedge \mathrm{Q}(x)) \\ \exists x\ (\mathrm{P}(x) \Rightarrow \mathrm{R}(x)) \\ \hline \therefore \exists x\ (\mathrm{Q}(x) \wedge \mathrm{R}(x)) \end{array}$$

Proof.

(1)	$\exists x\ (\mathrm{P}(x) \Rightarrow \mathrm{R}(x))$	hypothesis
(2)	$\mathrm{P}(c) \Rightarrow \mathrm{R}(c)$ for a particular c	existential instantiation, (1)
(3)	$\forall x\ (\mathrm{P}(x) \wedge \mathrm{Q}(x))$	hypothesis
(4)	$\mathrm{P}(c) \wedge \mathrm{Q}(c)$	universal instantiation, (3)
(5)	$\mathrm{Q}(c)$	simplification, (4)
(6)	$\mathrm{P}(c)$	simplification, (4)
(7)	$\mathrm{R}(c)$	modus ponens, (2) and (6)
(8)	$\mathrm{Q}(c) \wedge \mathrm{R}(c)$	conjunction, (5) and (7)
(9)	$\exists x\ (\mathrm{Q}(x) \wedge \mathrm{R}(x))$	existential generalization, (8)

2.2 Axioms and Formal Mathematical Theories

The rules of inference developed above tell us which deductive steps are valid because of their logical form. To develop a particular area of mathematics, however, we also need starting statements about the objects we intend to study. A *formal mathematical theory* consists of a specified language, logical principles and rules of inference, and a collection of *axioms*. An axiom is a statement assumed throughout the theory rather than proved from earlier statements in that theory.

Axioms should be distinguished from hypotheses and definitions. An axiom is a standing assumption of the entire theory, while a hypothesis is usually a temporary assumption made within a particular argument. A definition assigns meaning to a new term or symbol; it does not, by itself, guarantee that an object having the stated properties exists.

A *theorem* is a statement that can be deduced from the axioms using the accepted rules of inference. For example, if P and $P \Rightarrow Q$ are axioms or previously proved theorems, then modus ponens allows us to deduce Q. Once proved, Q may itself be used in later arguments. In a completely formal proof, every such step is written down and justified.

Calling a statement an axiom does not mean that it is self-evident or true in every possible mathematical setting. It describes the role that the statement plays within a particular theory. Different choices of axioms can lead to different theories and therefore to different collections of theorems.

Ordinary mathematical proofs do not include every elementary formal step. Instead, we write in mathematical English, make the important deductions explicit, and suppress routine applications of logic. Such proofs are nevertheless intended to be formalizable: in principle, the omitted steps could be supplied within an appropriate axiomatic theory. In the next chapter we will briefly examine ZFC, the standard axiomatic framework underlying most ordinary work with sets.

2.3 Proving Mathematical Theorems

Most mathematical statements are implications of the form P ⇒ Q, where P or Q might be compound propositions. In this section we will look at some methods for proving implications. Note that we are using some definitions and results from other areas of mathematics that we have not explicitly stated. We will not allow ourselves to make those kinds of assumptions once we begin our study of set theory.

2.3.1 Direct Proofs

A direct proof of the implication P ⇒ Q assumes that the hypothesis P is true, then uses definitions, previously proved theorems, and the rules of inference to deduce that the conclusion Q must also be true.

Example 2.8.

Prove the following theorem:

> **Theorem 2.9.**
> *If n is an even integer, then n^2 is even.*

Proof.

(1)	n is an even integer	hypothesis
(2)	$n = 2k$ for some integer k	definition, (1)
(3)	$n^2 = (2k)^2$	definition, (2)
(4)	$n^2 = 4k^2$	algebra, (3)
(5)	$n^2 = 2(2k^2)$	algebra, (4)
(6)	n^2 is even	definition, (5)

Although the two-column proof above is valid, it is not written in a format that mathematicians usually find pleasing. We prefer proofs that are written using complete sentences, paragraphs, and correct grammar and punctuation. Here is a better presentation for the previous proof.

Example 2.10.

Prove Theorem 2.9 in plain language.

Proof.

Let n be an even integer. By definition, $n = 2k$ for some integer k. Squaring both sides yields $n^2 = 4k^2 = 2(2k^2)$. Since $2k^2$ is an integer, the quantity $n^2 = 2(2k^2)$ is even as desired.

Example 2.11.

Prove the following theorem:

Theorem 2.12.
The square of an odd integer is odd.

Proof.

Let n be an odd integer. By definition, $n = 2k + 1$ for some integer k. Squaring both sides yields $n^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$. Since $2k^2 + 2k$ is an integer, the quantity $n^2 = 2(2k^2 + 2k) + 1$ is odd as desired.

2.3.2 Proving the Contrapositive

As you have shown in a homework exercise, the contrapositive of an implication is logically equivalent to the implication. Sometimes it is easier to prove the contrapositive of the implication we are interested in, as we will show here.

Example 2.13.
Prove the following theorem:

Theorem 2.14.
If n is an integer and n^2 is even, then n is even.

Proof.

We prove the contrapositive: if n is not even, then n^2 is not even. Let n be an integer that is not even. Every integer is exactly one of even or odd, so n is odd. By Theorem 2.12, n^2 is odd. Since no integer is both even and odd, n^2 is not even. This proves the contrapositive and therefore the original implication.

Example 2.15.
Prove the following theorem:

Theorem 2.16. *If n is an integer and n^2 is odd, then n is odd.*

Proof.

We prove the contrapositive: if n is not odd, then n^2 is not odd. Let n be an integer that is not odd. Every integer is exactly one of even or odd, so n is even. By Theorem 2.9, n^2 is even. Since no integer is both even and odd, n^2 is not odd. This proves the contrapositive and therefore the original implication.

2.3.3 Proof by Contradiction

A proof by contradiction, or *reductio ad absurdum*, begins by assuming that the result we are trying to prove is false, then deriving a contradiction from that assumption. It is easy to confuse this with a direct proof of the contrapositive, which makes different beginning assumptions. To prove the implication $P \Rightarrow Q$ by proving the contrapositive, you assume $\neg Q$ and use that to prove $\neg P$. A proof by contradiction assumes $P \wedge \neg Q$ and derives a contradiction $R \wedge \neg R$. Here is a classic example of a proof by contradiction.

Example 2.17.
Prove the following theorem:

Theorem 2.18.
There is no rational number x such that $x^2 = 2$.

Proof.

Suppose to the contrary that there is a rational number x with $x^2 = 2$. Since x is rational, we may write $x = p/q$, where $p, q \in \mathbb{Z}$, $q \neq 0$, and $\gcd(p, q) = 1$. Since $x = p/q$ and $x^2 = 2$, we have $p^2/q^2 = 2$, or:

$$p^2 = 2q^2 \tag{2.1}$$

Since q^2 is an integer, $p^2 = 2q^2$ is even. Now p is an integer and p^2 is even, so Theorem 2.14 implies that p is even. This allows us to write $p = 2k$ for some integer k. We plug this into Equation (2.1) to see that $4k^2 = 2q^2$, so $q^2 = 2k^2$. Now q is an integer and q^2 is even, so another application of Theorem 2.14 implies that q is even. We now know that both p and q are even, so 2 divides both of them. This contradicts our choice of p and q, so our assumption that x is rational must be incorrect.

2.3.4 Proof by Cases

To prove the implication $P \Rightarrow Q$ by cases, we make use of the following argument:

$$\begin{array}{c} P \Rightarrow (R \vee S) \\ R \Rightarrow Q \\ S \Rightarrow Q \\ \hline \therefore P \Rightarrow Q \end{array}$$

You will be asked to show that this argument is valid in the homework.

Here is a very simple example of a proof by cases. We make use of some of the order properties of the real numbers. In particular, we know that multiplying both sides of an inequality by a positive number preserves the inequality, and multiplying both sides of an inequality by a negative number reverses the inequality.

Example 2.19.

Prove the following theorem:

> **Theorem 2.20.**
> *If x is a real number, then $x^2 \geq 0$.*

Proof.

If x is a real number, then either $x \geq 0$ or $x < 0$.

Case 1. If $x \geq 0$, then we may multiply both sides of this inequality by x to see that $x^2 \geq x \cdot 0 = 0$.

Case 2. If $x < 0$, then multiplying both sides of the inequality by x reverses the inequality, yielding $x^2 > x \cdot 0 = 0$. Since $x^2 > 0$ we have $x^2 \geq 0$.

In either case we have $x^2 \geq 0$, so the result is true.

2.4 Proving Quantified Statements

2.4.1 Universally Quantified Statements

On the simplest level, a universally quantified statement has the form $\forall x\, P(x)$. To prove it, let x be an arbitrary element of the required domain and prove $P(x)$ without using any special property of x beyond its membership in that domain. It is frequently helpful to think of such a statement as an implication whose hypothesis is that x belongs to the required domain and whose conclusion is $P(x)$. Since universally quantified statements can be viewed as implications, we can use the same proof techniques.

2.4.2 Existentially Quantified Statements

Existentially quantified statements are significantly different than implications and universally quantified statements. To prove the statement $\exists x, P(x)$ we must show that there is at least one x in the domain that satisfies $P(x)$. One way to do this is simply to find a single particular example.

Example 2.21.

Prove the following theorem:

Theorem 2.22.
There is a positive integer N *such that* N, $N+1$, $N+2$, $N+3$, *and* $N+4$ *are all composite (not prime).*

Proof.

Let $N = 6! + 2$. Clearly, N is divisible by 2. Likewise, $N + 1 = 6! + 3$ is divisible by 3, $N+2 = 6!+4$ is divisible by 4, $N+3 = 6!+5$ is divisible by 5, and $N+4 = 6!+6$ is divisible by 6.

There are some instances where it is easier to show that some x satisfies $\mathrm{P}(x)$ without actually finding a particular x that works. Here is an elementary example that works by demonstrating that one of two possible choices must work, without determining which it is.

Example 2.23.
Prove the following theorem:

Theorem 2.24.
There exist irrational numbers a and b so that a^b is rational.

Proof.

First note that $\sqrt{2}$ is irrational, but

$$\left(\sqrt{2}^{\sqrt{2}}\right)^{\sqrt{2}} = \sqrt{2}^{\sqrt{2}\sqrt{2}} = \sqrt{2}^{\,2} = 2$$

is rational. Now it is certainly true that $\sqrt{2}^{\sqrt{2}}$ is either rational or irrational. If $\sqrt{2}^{\sqrt{2}}$ is rational, then we set $a = \sqrt{2}$ and $b = \sqrt{2}$ to find the example we need. If $\sqrt{2}^{\sqrt{2}}$ is irrational, then we set $a = \sqrt{2}^{\sqrt{2}}$ and $b = \sqrt{2}$ to find the example we need. Either way, there are irrational numbers a and b such that a^b is rational.

2.4.3 Counterexamples

An example showing that a universally quantified statement is false is called a *counterexample* to that statement.

Example 2.25.
Disprove the following proposition:

Conjecture 2.26.
Every integer is even.

It probably seems clear to you that this statement is false, but how could you prove that? To prove that a proposition is false we must prove that its negation is true, so we first find the negation of the original statement:

Some integer is not even.

We have transformed our original problem (proving that a statement is false) into something more familiar (proving that a statement is true). How do we prove the second statement is true? As noted above, one way to prove an existentially quantified statement is simply to find the object it says exists. In this case, we must find an integer that is not even. For instance, the integer $x = 1$ is not even. Note that you shouldn't just say that there are such examples, you should give a particular one that your reader can check.

2.5 Mathematical Induction

There is another type of proof that is frequently used to prove statements about the natural numbers, i.e., the numbers $1, 2, 3, \ldots$.

> **Theorem 2.27** (The Principle of Mathematical Induction).
> *Let* $\mathrm{P}(n)$ *be a statement about the natural number* n*. If* $\mathrm{P}(1)$ *is true and if* $\mathrm{P}(k) \implies \mathrm{P}(k+1)$ *for every natural number* k*, then* $\mathrm{P}(n)$ *is true for every natural number* n*.*

All proofs by induction use the following basic outline. To prove $\mathrm{P}(n)$ for all natural numbers n,

1. Base Case: Show that $\mathrm{P}(1)$ is true.
2. Inductive Step: Prove the implication $\mathrm{P}(k) \implies \mathrm{P}(k+1)$ for all $k \in \mathbb{N}$. Note: in proving that $\mathrm{P}(k) \implies \mathrm{P}(k+1)$, the assumption that $\mathrm{P}(k)$ is true is often referred to as the inductive hypothesis.
3. It is considered good form to clearly state that the statement $\mathrm{P}(n)$ is now true for all natural numbers n by induction.

Example 2.28.

Prove the following theorem:

> **Theorem 2.29.**
> *For each natural number n, $\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$.*

Proof.

If $n = 1$, then

$$\sum_{i=1}^{n} i = \sum_{i=1}^{1} i = 1 = \frac{1(2)}{2} = \frac{n(n+1)}{2}.$$

So, the formula holds for $n = 1$. Now suppose that the formula holds for some $k \in \mathbb{N}$. In other words, we are assuming that $\sum_{i=1}^{k} i = \frac{k(k+1)}{2}$. We want to show that the formula must hold for $n = k + 1$. Using our assumption, we compute:

$$\begin{aligned}\sum_{i=1}^{k+1} i &= \left(\sum_{i=1}^{k} i\right) + (k+1)\\ &= \frac{k(k+1)}{2} + (k+1)\\ &= \frac{k(k+1)}{2} + \frac{2(k+1)}{2}\\ &= \frac{(k+1)(k+2)}{2}\end{aligned}$$

Hence the formula holds for $n = k + 1$. By induction, the formula must hold for all $n \in \mathbb{N}$.

Chapter 3

Set Theory

3.1 What is a Set?

3.1.1 Naive Set Theory

Early work with sets often used *unrestricted comprehension*: any condition was assumed to determine a set of all objects satisfying that condition. Russell's paradox shows that this principle is inconsistent. Modern axiomatic set theories replace it with restricted principles, such as forming a subset of an already specified set. In this course we use informal set language, but every set-builder description should have a stated ambient set.

3.1.2 Russell's Paradox

Example 3.1.

The Barber's Paradox. A certain small town has one barber, who is a man and shaves exactly those men in town who do not shave themselves. Does the barber shave himself?

Russell considered this sort of thing more formally by constructing the (purported) set

$$\mathrm{R} = \{x \mid x \notin x\}.$$

If $\mathrm{R} \in \mathrm{R}$, then its defining condition gives $\mathrm{R} \notin \mathrm{R}$; if $\mathrm{R} \notin \mathrm{R}$, then its defining condition gives $\mathrm{R} \in \mathrm{R}$. The lesson is not merely that self-membership should be forbidden. Rather, the collection of all objects satisfying an arbitrary condition need not be a set. Standard axiomatic systems avoid the contradiction by restricting comprehension, as described below.

3.1.3 A Glimpse of ZFC

The most commonly used formal foundation for ordinary mathematics is *Zermelo–Fraenkel set theory with the Axiom of Choice*, usually abbreviated *ZFC*. It is not the only possible foundation, but it is the standard one. ZFC uses first-order logic with equality and one basic relation, membership $\in$. Every variable in its formal language ranges over sets. Numbers, ordered pairs, functions, and other mathematical objects can all be represented using sets, although we will continue to treat them as the familiar objects that they are. For example, one common set-theoretic representation of an ordered pair is

$$(a, b) = \{\{a\}, \{a, b\}\}.$$

The *Axiom of Extensionality* formalizes our criterion for equality of sets:

$$\forall A\, \forall B \left[\left(\forall x\, (x \in A \Leftrightarrow x \in B) \right) \Rightarrow A = B \right].$$

In words, if two sets have exactly the same elements, then they are equal. Consequently, a set is completely determined by its elements.

The principle that most directly addresses Russell's paradox is the *Separation schema*. For every property $\varphi(x)$ that can be expressed in the language of set theory, Separation supplies an axiom of the form

$$\forall A\, \exists B\, \forall x \left[x \in B \Leftrightarrow \left(x \in A \wedge \varphi(x) \right) \right].$$

Thus, from an already existing set A, we may form the subset

$$B = \{x \in A \mid \varphi(x)\}.$$

The word *schema* means that this is actually a pattern giving one axiom for each allowable formula φ. Separation does not say that $\{x \mid \varphi(x)\}$ exists without an ambient set. In particular, it does not provide Russell's purported set $\{x \mid x \notin x\}$.

The other axioms guarantee the existence of the sets needed for familiar constructions. Pairing gives sets such as $\{a, b\}$; Union and Power Set justify the formation of unions and power sets; and Infinity guarantees the existence of an infinite set. The Replacement schema allows the outputs of a definable functional rule, when applied to the elements of a set, to be collected into another set. Foundation rules out membership loops such as $x \in x$. Finally, the Axiom of Choice says that, given a set of nonempty sets, there is a function choosing one element from each of them. ZF consists of the preceding set-theoretic principles without Choice; ZFC means ZF together with Choice. Equivalent presentations sometimes package a few of these principles differently.

These axioms are what justify the constructions that we use informally. For example, Pairing guarantees that $\{A, B\}$ exists, and the Union axiom then guarantees the existence of $A \cup B$. The Power Set axiom guarantees the existence of $\mathcal{P}(A)$. A definition describes the intended set, while the axioms ensure that such a set actually exists.

We will not derive our results directly from the axioms of ZFC. The purpose of this discussion is only to show the formal framework underneath our informal work. In particular, ZFC has no set containing every set. When we later use a universal set U, it will mean a fixed ambient set for the problem under consideration, not a set containing all mathematical objects.

3.2 Elements and Subsets

3.2.1 Elements

We will think of a set as a collection of objects, called its *elements*. These objects can be any objects we are interested in discussing. Any roster or description must determine membership unambiguously. In particular, a set-builder description should specify an ambient set and a condition that determines membership. We consider two sets equal if they contain exactly the same elements. An object is either an element of a set or it is not; elements have no order and repetition does not create a new element.

One way to indicate a set is to simply list all of the elements between set braces; i.e., between the symbols $\{$ and $\}$. The set of positive integers less than 3 is $\{1, 2\}$. We will find it useful to have a special notation for the set which has no elements. To this end, we let the symbol $\varnothing$ denote the set $\{\}$ (i.e. $\varnothing = \{\}$).

Example 3.2.

Let

$$A = \{1, 2, 4, 8\}, \quad B = \{8, 2, 1, 4\}, \quad C = \{1, 2, 1, 4, 8\}, \quad D = \{1, 2, 3, 4, 8\}.$$

Of these sets, $A = B = C$ because they each contain the same elements.

It doesn't matter that we listed the elements in a different order when we defined A than when we defined B, or that we listed 1 as an element of C more than once.

The set D is different from the others since it contains the element 3 and the other sets do not.

Example 3.3.

Let $X = \{1, 2, \{2\}\}$. Note that this set contains some elements that are numbers and some elements that are sets of numbers. We consider those different objects.

For example 1 is an element of X, but $\{1\}$ is not an element of X. On the other hand, both 2 and $\{2\}$ are elements of X.

We indicate that the object x is an element of the set A by writing $x \in A$. We write $x \notin A$ if x is not an element of A.

Listing all elements works well for a small finite set. For larger sets we use *set-builder notation*

$$\{x \in U \mid P(x)\},$$

read "the set of all x in U such that $P(x)$." The ambient set U is part of the definition. For example, $\{x \in \mathbb{R} \mid x^2 - 2 = 0\}$ is a two-element set, whereas $\{x \in \mathbb{Q} \mid x^2 - 2 = 0\} = \varnothing$.

There are also a few sets that are useful enough to deserve their own special notation. In particular, $\mathbb{N}$ denotes the set of natural numbers, $\mathbb{Z}$ the set of integers, $\mathbb{Q}$ the set of rational numbers, $\mathbb{R}$ the set of real numbers, and $\mathbb{C}$ the set of complex numbers.

3.2.2 Subsets

Definition 3.4.
Given two sets A and B, we say that A is a *subset* of B, or that A is contained in B, if every element of A is also an element of B. We write $A \subseteq B$.

Since every natural number is also an integer and every integer is a real number, $\mathbb{N} \subsetneq \mathbb{Z} \subsetneq \mathbb{R}$.

Example 3.5.

Define the sets $A = \{1, 2, 3\}$, $\quad B = \{1, 2, \{3\}, 4\}$, $\quad C = \{1, 2, 3, \{4\}\}$. Then A is a subset of C because each element of A is also an element of C. Note that A is not a subset of B because 3 is an element of A, but not an element of B–remember that 3 and $\{3\}$ are different objects.

Given any set A, it is certainly true that every element of A is an element of A. In other words, every set is a subset of itself. Note also that the empty set is a subset of every set A since there are no elements of the empty set which are not in A.

Definition 3.6.
A subset B of A is a *proper subset* of A if $B \neq A$. We write $B \subsetneq A$. In particular, $\varnothing \subsetneq A$ whenever A is nonempty.

Theorem 3.7.
If $A \subseteq B$ *and* $B \subseteq C$, *then* $A \subseteq C$.

Proof.

Suppose that $a \in A$. Since $A \subseteq B$ it follows that $a \in B$. Now $B \subseteq C$, so we may say that $a \in C$ as desired.

3.2.3 Universal Sets

In many instances, all of the sets we may be interested in are subsets of some particular set U. In this case we say that U is a *universal set*, or that U is the *universe*. For example, all of the functions we study in single variable calculus have domains and ranges that are subsets of the universal set $\mathbb{R}$.

3.2.4 Equality of Sets

We say that two sets are equal if they contain exactly the same elements. In other words, $A = B$ means that every element of A is an element of B and every element of B is an element of A. A good test for this is the following one:

Theorem 3.8.
Given any two sets A *and* B, $A = B$ *if and only if* $A \subseteq B$ *and* $B \subseteq A$.

This theorem will become one of our most valuable tools for showing that two sets are equal.

3.3 Operations on Sets

3.3.1 Union and Intersection

Definition 3.9.
The *union* of two sets A and B is the set:

$$A \cup B = \{x \mid (x \in A) \vee (x \in B)\}$$

Definition 3.10.
The *intersection* of A and B is the set:

$$A \cap B = \{x \mid (x \in A) \wedge (x \in B)\}$$

Example 3.11.

Define $A = \{1, 2, 3, 4, 5\}$ and $B = \{2, 4, 6, 8, 10\}$. Then $A \cup B = \{1, 2, 3, 4, 5, 6, 8, 10\}$ and $A \cap B = \{2, 4\}$.

Theorem 3.12.
Let A, B, and C be any sets in the universal set U. Then:

1. *(Commutative Laws)*
   (a) $A \cup B = B \cup A$
   (b) $A \cap B = B \cap A$
2. *(Associative Laws)*
   (a) $A \cup (B \cup C) = (A \cup B) \cup C$
   (b) $A \cap (B \cap C) = (A \cap B) \cap C$
3. *(Distributive Laws)*
   (a) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
   (b) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
4. *(Idempotence)*
   (a) $A \cup A = A$
   (b) $A \cap A = A$
5. *(Identities)*
   (a) $A \cup \varnothing = A$
   (b) $A \cap U = A$

Proof.

We prove some of these properties here and leave the remainder for the exercises.

Commutative Laws. Note that $P \vee Q$ is equivalent to $Q \vee P$, so it follows immediately from our definition that $A \cup B = B \cup A$. Similarly we have $A \cap B = B \cap A$ since $P \wedge Q$ is equivalent to $Q \wedge P$.

Associative Laws. We prove the Associative Law for unions here. Rather than applying Theorem 3.8, we show that $x \in A \cup (B \cup C)$ is equivalent to $x \in (A \cup B) \cup C$ using the associativity of disjunction at $(*)$.

$$
\begin{aligned}
x \in A \cup (B \cup C) &\Leftrightarrow (x \in A) \vee (x \in B \cup C) \\
&\Leftrightarrow (x \in A) \vee \big((x \in B) \vee (x \in C)\big) \\
&\Leftrightarrow \big((x \in A) \vee (x \in B)\big) \vee (x \in C) \qquad (*) \\
&\Leftrightarrow (x \in A \cup B) \vee (x \in C) \\
&\Leftrightarrow x \in (A \cup B) \cup C
\end{aligned}
$$

Distributive Laws. We show that union distributes over intersection using the fact that disjunction distributes over conjunction at (**).

$$
\begin{aligned}
x \in A \cup (B \cap C) &\Leftrightarrow (x \in A) \vee (x \in B \cap C) \\
&\Leftrightarrow (x \in A) \vee \big((x \in B) \wedge (x \in C)\big) \\
&\Leftrightarrow \big((x \in A) \vee (x \in B)\big) \wedge \big((x \in A) \vee (x \in C)\big) \qquad (**) \\
&\Leftrightarrow (x \in A \cup B) \wedge (x \in A \cup C) \\
&\Leftrightarrow x \in (A \cup B) \cap (A \cup C)
\end{aligned}
$$

Idempotence. We prove that $A \cap A = A$. This time we make use of Theorem 3.8. Assume first that $x \in A \cap A$, then by definition $x \in A$ and $x \in A$. From this we may deduce that $x \in A$. Since this is true for every $x \in A \cap A$, we have shown that $A \cap A \subseteq A$. Now assume that $y \in A$, then we have $y \in A$ and $y \in A$. It follows that $y \in A \cap A$ for every $y \in A$, so $A \subseteq A \cap A$. We now apply Theorem 3.8 to see that $A \cap A = A$.

Identities. We prove here that $A \cup \varnothing = A$. Assume first that $x \in A \cup \varnothing$, then $x \in A$ or $x \in \varnothing$. By definition we know that $x \notin \varnothing$, so it follows that $x \in A$. We have shown that $A \cup \varnothing \subseteq A$. Now suppose that $y \in A$, then we may deduce that $y \in A$ or $y \in \varnothing$. By definition it follows that $y \in A \cup \varnothing$ and we have shown that $A \subseteq A \cup \varnothing$. We may now apply Theorem 3.8 to see that $A \cup \varnothing = A$ as desired.

3.3.2 Complements

Definition 3.13.
The *relative complement* of B in A (also called the set difference) is the set:

$$A \setminus B = \{a \mid a \in A \text{ and } a \notin B\}$$

In a given universe U, we may also define the *complement* of a set A to be the set:

$$A' = U \setminus A$$

Theorem 3.14 (De Morgan's Laws).
Let A, B, *and* C *be sets. Then:*

(i) $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$, *and*

(ii) $A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)$.

Proofs.

(i)

$$\begin{aligned} x \in A \setminus (B \cup C) &\Leftrightarrow (x \in A) \wedge (x \notin B \cup C) \\ &\Leftrightarrow (x \in A) \wedge \neg(x \in B \vee x \in C) \\ &\Leftrightarrow (x \in A) \wedge \big(\neg(x \in B) \wedge \neg(x \in C)\big) \\ &\Leftrightarrow (x \in A) \wedge (x \notin B) \wedge (x \notin C) \\ &\Leftrightarrow \big((x \in A) \wedge (x \notin B)\big) \wedge \big((x \in A) \wedge (x \notin C)\big) \\ &\Leftrightarrow (x \in A \setminus B) \wedge (x \in A \setminus C) \\ &\Leftrightarrow x \in (A \setminus B) \cap (A \setminus C) \end{aligned}$$

(ii)

$$\begin{aligned}
x \in A \setminus (B \cap C) &\Leftrightarrow (x \in A) \wedge (x \notin B \cap C) \\
&\Leftrightarrow (x \in A) \wedge \neg(x \in B \wedge x \in C) \\
&\Leftrightarrow (x \in A) \wedge \big(\neg(x \in B) \vee \neg(x \in C)\big) \\
&\Leftrightarrow (x \in A) \wedge \big((x \notin B) \vee (x \notin C)\big) \\
&\Leftrightarrow \big((x \in A) \wedge (x \notin B)\big) \vee \big((x \in A) \wedge (x \notin C)\big) \\
&\Leftrightarrow (x \in A \setminus B) \vee (x \in A \setminus C) \\
&\Leftrightarrow x \in (A \setminus B) \cup (A \setminus C)
\end{aligned}$$

3.3.3 Cartesian Products

Definition 3.15.
The *(Cartesian) product* of two sets A and B is the set:

$$A \times B = \{(a, b) \mid a \in A \text{ and } b \in B\},$$

where (a, b) denotes an ordered pair, not an interval.

Ordered pairs satisfy $(a, b) = (a', b')$ if and only if $a = a'$ and $b = b'$.

Note that $A \times B$ is simply the set of all ordered pairs with first coordinates in A and second coordinates in B. For example, the Cartesian plane used in Calculus is the set $\mathbb{R} \times \mathbb{R}$.

Theorem 3.16.
For any sets A, B, *and* C*:*

(i) $(A \cup B) \times C = (A \times C) \cup (B \times C)$

(ii) $(A \cap B) \times C = (A \times C) \cap (B \times C)$

(iii) $(A \setminus B) \times C = (A \times C) \setminus (B \times C)$

Proof.

We leave parts (ii) and (iii) as exercises.

(i) First suppose that $(x, y) \in (A \cup B) \times C$. By definition we have $x \in A \cup B$ and $y \in C$. Since $x \in A \cup B$, either $x \in A$ or $x \in B$. If $x \in A$, then we have $x \in A$ and $y \in C$, so $(x, y) \in A \times C$. If $x \in B$, then we have $x \in B$ and $y \in C$, so $(x, y) \in B \times C$. We can now say that $(x, y) \in A \times C$ or $(x, y) \in B \times C$, so $(x, y) \in (A \times C) \cup (B \times C)$. Hence $(A \cup B) \times C \subseteq (A \times C) \cup (B \times C)$.

To see that the converse is true, suppose that $(x, y) \in (A \times C) \cup (B \times C)$. Either $(x, y) \in A \times C$ or $(x, y) \in B \times C$. If $(x, y) \in A \times C$, then $x \in A$. Since $A \subseteq A \cup B$, it follows that $x \in A \cup B$; together with $y \in C$, this gives $(x, y) \in (A \cup B) \times C$. The analogous argument applies when $(x, y) \in B \times C$. Hence $(A \times C) \cup (B \times C) \subseteq (A \cup B) \times C$. Therefore, $(A \cup B) \times C = (A \times C) \cup (B \times C)$ as desired.

The products $A \times B$ and $B \times A$ need not be equal because their coordinates play different roles. They are equal in special cases, for example when $A = B$ or when at least one factor is empty. The proof of the following theorem involves making obvious changes to the previous proof and checking that all of the details still work.

Theorem 3.17.
For any sets A, B, *and* C:

(i) $A \times (B \cup C) = (A \times B) \cup (A \times C)$

(ii) $A \times (B \cap C) = (A \times B) \cap (A \times C)$

(iii) $A \times (B \setminus C) = (A \times B) \setminus (A \times C)$

It may be tempting to assume that we could combine these two theorems somehow and obtain statements like $(A \times B) \setminus (C \times D) = (A \setminus C) \times (B \setminus D)$. This statement is generally false, however.

Example 3.18.

Let $A = \{1, 2, 3\}$, $B = \{5, 6\}$, $C = \{1, 2\}$, and $D = \{6\}$. Then:

$$(A \times B) \setminus (C \times D) = \{(1, 5), (2, 5), (3, 5), (3, 6)\}$$

but

$$(A \setminus C) \times (B \setminus D) = \{(3, 5)\}$$

3.4 Collections of Sets

Some of the structures used in pure mathematics require the use of sets whose elements are other sets. It is customary, though not necessary, to refer to these kinds of sets as collections of sets.

3.4.1 The Power Set of a Set

Definition 3.19.
Let A be any set. The *power set* of A is $\mathcal{P}(A) = \{B \mid B \subseteq A\}$.

In words, the power set of A is the set whose elements are the subsets of A. Note that for any set A we have $\varnothing \in \mathcal{P}(A)$ and $A \in \mathcal{P}(A)$, so $\mathcal{P}(A)$ is nonempty for every set A. The power set of A is frequently denoted 2^A.

Theorem 3.20.
For any sets A *and* B, $A \subseteq B$ *if and only if* $\mathcal{P}(A) \subseteq \mathcal{P}(B)$.

Proof.

First assume that $A \subseteq B$ and let $X \in \mathcal{P}(A)$. By definition $X \subseteq A$, so Theorem 3.7 implies that $X \subseteq B$. Hence $X \in \mathcal{P}(B)$ and $\mathcal{P}(A) \subseteq \mathcal{P}(B)$. Conversely, assume that $\mathcal{P}(A) \subseteq \mathcal{P}(B)$. Since $A \in \mathcal{P}(A)$, it follows that $A \in \mathcal{P}(B)$, and therefore $A \subseteq B$ by the definition of the power set.

Theorem 3.21.
For any sets A *and* B, $\mathcal{P}(A \cap B) = \mathcal{P}(A) \cap \mathcal{P}(B)$.

Proof.

First note that $A \cap B \subseteq A$ by an exercise in the homework. So, Theorem 3.20 implies that $\mathcal{P}(A \cap B) \subseteq \mathcal{P}(A)$. The same reasoning shows that $\mathcal{P}(A \cap B) \subseteq \mathcal{P}(B)$, so we have $\mathcal{P}(A \cap B) \subseteq \mathcal{P}(A) \cap \mathcal{P}(B)$.

Now suppose that $X \in \mathcal{P}(A) \cap \mathcal{P}(B)$. Then $X \in \mathcal{P}(A)$ and $X \in \mathcal{P}(B)$. By definition $X \subseteq A$ and $X \subseteq B$, so $X \subseteq A \cap B$ by a problem from the homework. It follows that $\mathcal{P}(A \cap B) \supseteq \mathcal{P}(A) \cap \mathcal{P}(B)$, so $\mathcal{P}(A \cap B) = \mathcal{P}(A) \cap \mathcal{P}(B)$ as desired.

Chapter 4

Relations

Introduction

The first mathematical relation most students become aware of is the standard order on the natural numbers, which is later extended to the integers and finally the reals. A relation is a much more general concept, and by now you have probably worked with several others, perhaps without thinking about them as relations. In this chapter we define what we mean by a relation between sets, then define an equivalence relation on a set and use that idea to develop the set of rational numbers.

4.1 Relations

Definition 4.1.
Given two sets A and B, a *relation* from A to B is a subset of $A \times B$. The relations we will be most interested in are usually from a set A to itself, in which case we say that the relation is a relation on A. If S is a relation, we frequently use the notation xSy to indicate that the ordered pair (x, y) is in S.

Let's look at some examples of relations.

Example 4.2.

The usual ordering $<$ on $\mathbb{R}$ is a relation on $\mathbb{R}$. We don't usually think of this relation as a set of ordered pairs, but we could. Define a subset L of $\mathbb{R} \times \mathbb{R}$ by $L = \{(a, b) \mid b - a \text{ is positive}\}$. In other words, an ordered pair (a, b) is in the relation if $a < b$.

Example 4.3.

The set of points on a circle is another example of a relation on $\mathbb{R}$. For example, we could let $C = \{(x, y) \mid x^2 + y^2 = 1\}$.

Example 4.4.

A relation need not be something you've encountered before or that you necessarily have a use for. Let A denote the set of people in this room, and B the set of possible hair colors. One might define a relation H from A to B by: $H = \{(x, y) \mid \text{y is x's hair color}\}$.

Definition 4.5.
Let A be a set and let $\sim$ be a relation on A. We say that $\sim$ is:

1. *reflexive* if $a \sim a$ for every $a \in A$;
2. *symmetric* if $a \sim b$ implies $b \sim a$ for every $a, b \in A$;
3. *antisymmetric* if $a \sim b$ and $b \sim a$ imply $a = b$ for every $a, b \in A$.
4. *transitive* if $a \sim b$ and $b \sim c$ imply $a \sim c$ for every $a, b, c \in A$.

Note that symmetry and antisymmetry are not logical opposites, though the names may lead you to believe otherwise. It is possible for a relation to satisfy both of these properties, or to satisfy neither of them.

Example 4.6.

Let $<$ denote the relation "less than" on $\mathbb{R}$. Determine which of the properties listed above are satisfied by $<$.

- $<$ is not reflexive since $1 \not< 1$
- $<$ is not symmetric since $1 < 7$ and $7 \not< 1$
- $<$ is antisymmetric because the hypotheses $a < b$ and $b < a$ are never both satisfied
- $<$ is transitive since $a < b$ and $b < c$ implies that $a < c$

4.2 Partial Orders

Definition 4.7. A relation $\preceq$ on a set A is a *partial order* if it is reflexive, antisymmetric, and transitive. The pair $(A, \preceq)$ is a *partially ordered set*, or *poset*.

Two elements $a, b \in A$ are *comparable* if $a \preceq b$ or $b \preceq a$. A partial order in which every pair of elements is comparable is called a *total order*.

Example 4.8.

The usual relation $\leq$ is a total order on $\mathbb{R}$. It is reflexive because $a \leq a$; antisymmetric because $a \leq b$ and $b \leq a$ force $a = b$; and transitive because $a \leq b \leq c$ implies $a \leq c$.

Example 4.9.

For a fixed set S, inclusion $\subseteq$ is a partial order on $\mathcal{P}(S)$. It is usually not a total order: if $a, b \in S$ are distinct, then neither $\{a\} \subseteq \{b\}$ nor $\{b\} \subseteq \{a\}$. Thus some pairs can be incomparable.

4.3 Equivalence Relations

Consider the following question: Are $\pi/4$ and $25\pi/4$ measurements of the same angle? It's certainly true that when we place these angles in standard position they have the same terminal side, which means that we can treat them as the same much of the time. On the other hand, if we think of the angle as a rotation, these two angles are certainly different. If you're playing pin the tail on the donkey, being spun through an angle of $25\pi/4$ is going to make you dizzier than being spun through an angle of $\pi/4$. Intuitively, the terminal side of the angle in this setting tells us what direction we end up pointing in. The angle tells us how far we rotated to end up pointing in that direction. When we are only concerned with the terminal side of an angle, how do we tell when two numbers are measures for *equivalent* angles? You probably learned in a trigonometry or precalculus class that two measurements represent equivalent angles when they differ by an integer multiple of 2π. In other words, two angles α and β are said to be *equivalent* when there is an integer n such that $\alpha - \beta = 2\pi n$. In the following example we consider this relation between real numbers.

Example 4.10.

For $x, y \in \mathbb{R}$ we define $x \sim y$ if there is an integer n such that $x - y = 2\pi n$. Show that $\sim$ is reflexive, symmetric, and transitive.

- For any $x \in \mathbb{R}$ we have $x - x = 0 = 2\pi(0)$. Since $0 \in \mathbb{Z}$, $x \sim x$ and $\sim$ is reflexive.
- Suppose that $x \sim y$, then there is an $n \in \mathbb{Z}$ so that $x - y = 2\pi n$. It follows that $y - x = 2\pi(-n)$. Since $-n \in \mathbb{Z}$ we have $y \sim x$ and $\sim$ is symmetric.
- Let $x, y, z \in \mathbb{R}$; assume that $x \sim y$ and $y \sim z$. By definition there are integers $n, m \in \mathbb{Z}$ such that $x - y = 2\pi n$ and $y - z = 2\pi m$. It follows that

$$x - z = (x - y) + (y - z) = 2\pi n + 2\pi m = 2\pi(n + m).$$

Thus $x \sim z$, which shows that $\sim$ is transitive.

Definition 4.11.
A relation on a set A that is reflexive, symmetric, and transitive is said to be an *equivalence relation* on A.

4.3.1 Equivalence Classes

Definition 4.12.
Let A be a set and let $\sim$ be an equivalence relation on A. For any $a \in$ A the set $[a] = \{b \in \mathrm{A} \mid b \sim a\}$ is called the *equivalence class* of a.

Given a set A and an equivalence relation on A, the set of equivalence classes forms a *partition* of A. In other words, the equivalence classes are nonempty subsets of A, every element of A belongs to an equivalence class, and no two distinct equivalence classes intersect. We formalize this in the following theorem.

Theorem 4.13.
Let A *be a nonempty set and* $\sim$ *an equivalence relation on* A*. For each* $a \in$ A*, let*

$$[a] = \{b \in \mathrm{A} \mid b \sim a\}.$$

(i) For each $a \in \mathrm{A}$, $a \in [a]$.

(ii) If $[a] \neq [b]$*, then* $[a] \cap [b] = \varnothing$.

Proof.

We begin with part (i): Since $\sim$ is reflexive, $a \sim a$ for each $a \in$ A. By definition this implies that $a \in [a]$ and (i) is satisfied. For part (ii): Suppose that $[a] \cap [b] \neq \varnothing$ and let $c \in [a] \cap [b]$. Then $c \sim a$ and $c \sim b$. Since $c \sim a$ and $\sim$ is symmetric, it follows that $a \sim c$. Now we have $a \sim c$ and $c \sim b$, so $a \sim b$ by transitivity. Let $x \in [a]$, then $x \sim a$ by definition. From transitivity it follows that $x \sim b$, so $x \in [b]$. Hence $[a] \subseteq [b]$.

Since $a \sim b$, it must also be true that $b \sim a$ by symmetry. Let $y \in [b]$, then $y \sim b$ by definition. From transitivity it follows that $y \sim a$, so $y \in [a]$. Hence $[b] \subseteq [a]$. Since $[a] \subseteq [b]$ and $[b] \subseteq [a]$, $[a] = [b]$ by Theorem 3.8.

Example 4.14.

Consider the equivalence relation $\equiv_5$ on $\mathbb{Z}$ defined by: $a \equiv_5 b$ if there is an integer n so that $b - a = 5n$. There are five equivalence classes associated with this equivalence relation:

$$[0] = \{\dots, -10, -5, 0, 5, 10, \dots\}$$

$$[1] = \{\dots, -9, -4, 1, 6, 11, \dots\}$$

$$[2] = \{\dots, -8, -3, 2, 7, 12, \dots\}$$

$$[3] = \{\dots, -7, -2, 3, 8, 13, \dots\}$$

$$[4] = \{\dots, -6, -1, 4, 9, 14, \dots\}$$

The equivalence classes are sets of integers that have the same remainder when divided by 5.

In the previous example, note that we had several ways to refer to each equivalence class. For example $[1] = [16]$, so we may as well have used $[16]$ as a name for this equivalence class. In this context the numbers 1 and 16 (or any other member of the class) are called *representatives* of this equivalence class, and any representative can be used to name the equivalence class.

4.4 The Rational Numbers

In this section we use equivalence classes to construct the rational numbers. Assume that $\mathbb{N}$ and $\mathbb{Z}$ have their usual properties and recall that $0 \notin \mathbb{N}$ in this text. Begin with

$$\mathbb{F} = \mathbb{Z} \times \mathbb{N}.$$

Its elements are ordered pairs (a, b), not yet numerical quotients. We use a/b as a convenient *formal symbol* for the pair (a, b); the condition $b \in \mathbb{N}$ guarantees that its second coordinate is nonzero.

Note that a fraction is not quite the same thing as a rational number, for example $1/2$ and $3/6$ are different fractions that represent the same rational number. We define an equivalence relation $\cong$ on $\mathbb{F}$ that clarifies when two fractions represent the same rational as follows:

Definition 4.15.
Given two fractions a/b and c/d in $\mathbb{F}$, we say that $a/b \cong c/d$ if $ad = bc$.

Theorem 4.16.
The relation $\cong$ is an equivalence relation on $\mathbb{F}$.

Proof.

To see that $\cong$ is reflexive, note that for every $a/b \in \mathbb{F}$ we have $ab = ba$, so $a/b \cong a/b$.

Now suppose that $a/b \cong c/d$, then by definition $ad = bc$. Using the fact that multiplication is commutative we see that $cb = da$, so $c/d \cong a/b$ and $\cong$ is symmetric.

It remains to show that $\cong$ is transitive, which is a little bit more work. Assume that $a/b \cong c/d$ and $c/d \cong e/f$, then by definition we have $ad = bc$ and $cf = de$. Making use of our hypotheses, we perform some algebra, making sure that we do not try to divide by 0:

$$\begin{aligned}
ad &= bc \\
(ad)f &= (bc)f \\
(ad)f &= b(cf) \\
(ad)f &= b(de) \\
(af)d &= (be)d \\
af &= be \qquad \text{because } d \in \mathbb{N}, \text{ so } d \neq 0.
\end{aligned}$$

Now we have $a/b \cong e/f$ and $\cong$ is transitive as desired.

We now define the set of rationals to be the set of equivalence classes of fractions. In this construction the two fractions $1/2$ and $3/6$ do indeed represent the same rational number since they are representatives of the same equivalence class. This construction might be a bit unsatisfying in the sense that it tells us nothing about how we might add or multiply rational numbers. Even if we know how to add fractions, how would we add two equivalence classes of fractions? One possibility is that we might add two equivalence classes by adding their representatives, so for example we might try:

$$\left[\frac{1}{2}\right] \oplus \left[\frac{2}{3}\right] = \left[\frac{1(3)+2(2)}{2(3)}\right] = \left[\frac{5}{6}\right]$$

There is a potential problem with this, though. There are infinitely many choices of fractions representing each rational number. Are we sure that we arrive at the same result for all of those possible choices? The next theorem says that we do.

Theorem 4.17.
Suppose that $a/b \cong x/y$ and $c/d \cong w/z$, where a/b, c/d, x/y, and w/z are all in $\mathbb{F}$, then:

$$\frac{ad+bc}{bd} \cong \frac{xz+yw}{yz}.$$

Proof.

By hypothesis we have $ay = bx$ and $cz = dw$. We want to show that $(ad+bc)(yz) = (bd)(xz+yw)$. Using our hypotheses we have:

$$\begin{aligned}(ad+bc)(yz) &= (ad)(yz)+(bc)(yz)\\ &= (ay)(dz)+(cz)(by)\\ &= (bx)(dz)+(dw)(by)\\ &= (bd)(xz)+(bd)(yw)\\ &= (bd)(xz+yw).\end{aligned}$$

The proof of the following theorem is left as an exercise.

Theorem 4.18.
Suppose that $a/b \cong x/y$ and $c/d \cong w/z$, where a/b, c/d, x/y, and w/z are all in $\mathbb{F}$, then:

$$\frac{ac}{bd} \cong \frac{xw}{yz}.$$

Theorem 4.17 and Theorem 4.18 allow us to make the following definitions, where we may choose any representative from each equivalence class.

Definition 4.19.
Let $[a/b]$ and $[c/d]$ be rational numbers, then we define addition and multiplication by:

$$\left[\frac{a}{b}\right] \oplus \left[\frac{c}{d}\right] = \left[\frac{ad+bc}{bd}\right] \quad \text{and} \quad \left[\frac{a}{b}\right] \otimes \left[\frac{c}{d}\right] = \left[\frac{ac}{bd}\right].$$

Chapter 5

Functions

Introduction

As anybody who has taken a calculus class knows, functions are important in mathematics. In this chapter we will define a function between two sets as a special type of relation between those sets. This set-theoretic definition emerged from nineteenth- and early twentieth-century efforts to make the notions of function, relation, and set mathematically precise. These efforts became especially important once paradoxes exposed limitations in informal set reasoning.

5.1 Definition

You probably think of functions in several ways based on what you have seen in previous courses. Functions can be rules for computing things, some people think of them as machines (plug in one number and get out another), and of course we all know that a graph is only the graph of a function if it passes the vertical line test. Most of these ideas match the way that mathematicians thought of functions (and the way they worked with them) at some time or another. It wasn't until the late 19th century that the concept of a function was defined in terms of sets. We present such a definition here:

Definition 5.1.
Let A and B be sets. A *function* f from A to B is a relation from A to B with the additional property that for each $a \in A$ there is exactly one ordered pair (a, b) in f having a as a first coordinate. In this case, the set A is called the *domain* of f and the set B is called the *codomain* of f. If f is a function from A to B, we denote this by writing $f : A \to B$.

Do not confuse the codomain of a function with its range. The codomain of a function is the set that second coordinates must come from. The range is the subset of the codomain containing all of those second coordinates. Sometimes those sets are the same, but sometimes they are not.

Note that our definition requires that every element of the domain be the first coordinate of one, and only one, ordered pair in a function. It does not, however, require that each element of the codomain be a second coordinate, or that it be the second coordinate of only one ordered pair. Satisfying these requirements would make our function surjective or injective, respectively. We discuss these kinds of functions in Section 5.2.

Before going any further, let's consider a couple of examples.

Example 5.2.

Consider the sets $A = \{1, 2, 3, 4\}$ and $B = \{1, 3, 5, 7\}$. Define the following relations from A to B:

(i) $f = \{(1, 3), (2, 1), (3, 5), (4, 7)\}$

(ii) $g = \{(1, 1), (3, 3)\}$

(iii) $h = \{(1, 1), (2, 3), (3, 5), (4, 1)\}$

(iv) $j = \{(1, 1), (2, 3), (3, 5), (4, 7), (2, 7)\}$

(v) $\mathrm{Pat} = \{(1, 1), (2, 3), (3, 5), (4, 7), (1, 1)\}$

The relation f is certainly a function. Each element of A is a first coordinate of one and only one ordered pair.

The relation g is not a function because 2 and 4 are elements of A, but are not first coordinates of ordered pairs.

The relation h is a function. The fact that 1 is a second coordinate of two ordered pairs, or that 7 is not the second coordinate of any, do not violate our definition.

The relation j is not a function because 2 is the first coordinate of two distinct ordered pairs.

Finally, Pat is also a function. We make two comments here. First, the fact that the ordered pair $(1, 1)$ is listed twice does not violate our definition of a function. Each ordered pair is either in the relation or not; it cannot be two distinct elements of the relation. Second, while we will usually use letters like f or g to denote functions, and adhering to this convention makes life easier for us, there is no real requirement that we do so. We can name a relation, and thus also a function, in some other manner if there is a reason to do so.

The functions in the previous example are obviously constructed to fit the definition, but may not strike you as the kinds of things you think of as functions. What about the kinds of functions we are used to?

Example 5.3.

Define the following relations from $\mathbb{R}$ to $\mathbb{R}$.

(i) $f = \{(x, y) \mid x \in \mathbb{R} \text{ and } y = x^2\}$

(ii) $g = \{(x, y) \mid x \in \mathbb{R} \text{ and } x = y^2\}$

The relation f is a function (make sure you understand why). In fact, it is a function that should be familiar to you. The second coordinates are squares of the first coordinates, so this is the usual squaring function on $\mathbb{R}$.

The relation g is not a function from $\mathbb{R}$ to $\mathbb{R}$. Despite appearances, there are elements of $\mathbb{R}$ which are not first coordinates: -1 is one such element, but any negative number will do. This relation also violates our definition in another way. Both $(4, 2)$ and $(4, -2)$ satisfy the definition of g, so 4 is the first coordinate of more than one ordered pair in g. In fact, every positive real number is the first coordinate of two ordered pairs in g.

It probably seems unnatural to you to write the squaring function $f : \mathbb{R} \to \mathbb{R}$ as we did in Example 5.3. Wouldn't it be easier to write this function as $f(x) = x^2$, the same way we did in algebra or calculus classes? Once we know that f is a function, we can use this notation.

Suppose that $f : A \to B$ is a function. If $a \in A$ and $(a, b) \in f$, we say that b is the *value* of the function f at a and denote this by writing $b = f(a)$.

Now the notation $f(x) = x^2$ indicates that for each $x \in \mathbb{R}$, x^2 is the value of the function at x. Please be careful to distinguish between the name of the function f and an arbitrary value of the function $f(x)$.

Example 5.4.

Here are some examples of important types of functions. Assume each of the sets is nonempty.

(i) Let A be any set. The identity function $i : A \to A$ is defined by $i(a) = a$ for each $a \in A$.

(ii) Let A and B be sets and let $b_0 \in B$. The function $k : A \to B$ defined by $k(a) = b_0$ for all $a \in A$ is called a constant function.

(iii) Let A and B be any sets. The coordinate projections $\pi_A : A \times B \to A$ and $\pi_B : A \times B \to B$ are defined by $\pi_A(a, b) = a$ and $\pi_B(a, b) = b$ for each $(a, b) \in A \times B$.

(iv) Let A be any subset of $\mathbb{R}$. The characteristic function $\chi_A : \mathbb{R} \to \{0, 1\}$ of A is defined by

$$\chi_A(x) = \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{if } x \notin A \end{cases}$$

Note: try and sketch a graph of the characteristic function of $\mathbb{Q}$.

5.1.1 Images and Preimages

Definition 5.5. Let $f : A \to B$ be a function.

(i) If $S \subseteq A$, the *image* of S under f is

$$f[S] = \{f(s) \mid s \in S\} \subseteq B.$$

In particular, the range of f is $f[A]$.

(ii) If $T \subseteq B$, the *preimage* of T under f is

$$f^{-1}[T] = \{a \in A \mid f(a) \in T\}.$$

The notation $f^{-1}[T]$ denotes a set and is meaningful for *every* function f, whether or not f has an inverse function. It should not be confused with the value $f^{-1}(y)$ of an inverse function. For example, if $f : \mathbb{R} \to \mathbb{R}$ is $f(x) = x^2$, then

$$f^{-1}[\{4\}] = \{-2, 2\},$$

even though f has no inverse function from $\mathbb{R}$ to $\mathbb{R}$.

Images and preimages interact differently with set operations. For $S_1, S_2 \subseteq A$ and $T_1, T_2 \subseteq B$,

$$\begin{aligned} f[S_1 \cup S_2] &= f[S_1] \cup f[S_2], \\ f^{-1}[T_1 \cup T_2] &= f^{-1}[T_1] \cup f^{-1}[T_2], \\ f^{-1}[T_1 \cap T_2] &= f^{-1}[T_1] \cap f^{-1}[T_2], \\ f^{-1}[B \setminus T_1] &= A \setminus f^{-1}[T_1]. \end{aligned}$$

In general only $f[S_1 \cap S_2] \subseteq f[S_1] \cap f[S_2]$; equality holds whenever f is injective.

5.1.2 Binary Operations

Standard addition and multiplication on $\mathbb{R}$ are both examples of functions used to compute a single real number from two given real numbers. This kind of function is important enough to deserve its own name.

Definition 5.6.
A *binary operation* on a set X is a function $* : X \times X \to X$. If $*$ is a binary operation on X, we usually use the notation $x * y$ to denote the value $*(x, y)$.

Example 5.7.

The standard addition and multiplication operations on $\mathbb{N}$, $\mathbb{Z}$, or $\mathbb{R}$ are all examples of binary operations. Note that they are all *different* binary operations.

Example 5.8.

Subtraction is a binary operation on $\mathbb{Z}$ and on $\mathbb{R}$. It is not a binary operation on $\mathbb{N}$: the integer difference $7 - 12 = -5$ is defined, but the result is not in $\mathbb{N}$. Thus $\mathbb{N}$ is not closed under subtraction.

Definition 5.9. We say that a binary operation $*$ on a set X is:

- *commutative* if $x * y = y * x$ for all $x, y \in X$.
- *associative* if $(x * y) * z = x * (y * z)$ for all $x, y, z \in X$.

Addition and multiplication on $\mathbb{Z}$ (or on $\mathbb{N}$ or $\mathbb{R}$ for that matter) are commutative and associative, which you have probably known since your first algebra class. Subtraction on $\mathbb{Z}$ is not commutative since, for example, $4 - 2 \neq 2 - 4$. Subtraction on $\mathbb{Z}$ also fails to be associative since, for example, $(5-1)-3 = 1 \neq 7 = 5-(1-3)$.

Example 5.10.

Define the operation $*$ on $\mathbb{Z}$ by $a * b = (ab)^2$. We claim that $*$ is commutative. To see this, let a and b be arbitrary integers. Then:

$$\begin{aligned} a * b &= (ab)^2 \\ &= a^2b^2 \\ &= b^2a^2 \\ &= (ba)^2 \\ &= b * a \end{aligned}$$

We leave it as an exercise to determine whether or not $*$ is associative.

5.2 Injective, Surjective, and Bijective Functions

Definition 5.11.
A function $f : X \to Y$ is said to be:

- *injective* or *one-to-one* if for all $x, y \in X$, if $f(x) = f(y)$, then $x = y$.
- *surjective* or *onto* if for each $y \in Y$, there is an $x \in X$ such that $f(x) = y$.
- *bijective* or *a one-to-one correspondence* if f is both injective and surjective.

Example 5.12. Classify the functions from Example 5.4 as injective, surjective, both, or neither, under the assumptions stated below.

(i) Let A be nonempty and consider the identity function $i : A \to A$.

The identity function is bijective. Assume first that $a, b \in A$ with $i(a) = i(b)$. Since $i(a) = a$ and $i(b) = b$, this implies that $a = b$, so i is injective. To see that i is surjective, let c be any element of A. Then $i(c) = c$.

(ii) Assume that A and B each have more than one element and let $k : A \to B$ be the constant function $k(a) = b_0$.

The function is neither injective nor surjective. Choose $a_1 \neq a_2$ in A; then $k(a_1) = b_0 = k(a_2)$. If $b \neq b_0$ and $b \in B$, then $b \neq k(a)$ for every $a \in A$.

(iii) Assume that A and B each have more than one element and consider the coordinate projections $\pi_A : A \times B \to A$ and $\pi_B : A \times B \to B$.

Both projections are surjective but not injective. For example, given $a \in A$, choose $b_0 \in B$; then $\pi_A(a, b_0) = a$. If $b_1 \neq b_2$, then $(a, b_1) \neq (a, b_2)$ but $\pi_A(a, b_1) = a = \pi_A(a, b_2)$. The proof for π_B is similar.

(iv) Consider the characteristic function $\chi_A : \mathbb{R} \to \{0, 1\}$.

It is surjective exactly when both A and $\mathbb{R} \setminus A$ are nonempty: one set supplies an input with value 1, and the other supplies an input with value 0. It is never injective: the three distinct inputs 0, 1, and 2 produce only the two possible values 0 and 1, so at least two of those inputs have the same value.

5.3 Compositions of Functions

Definition 5.13.
Let $f : X \to Y$ and $g : Y \to Z$ be functions. The *composition* is the function from X to Z defined by $(g \circ f)(x) = g(f(x))$. In other words, to find $(g \circ f)(x)$, first find $f(x)$, then plug the result into the function g. In terms of ordered pairs, the composition is

$$g \circ f = \{(x, z) \mid (x, y) \in f \text{ and } (y, z) \in g \text{ for some } y \in Y\}.$$

Theorem 5.14. *Let $f : X \to Y$ and $g : Y \to Z$ be functions.*

(i) If f and g are injective, then $g \circ f : X \to Z$ is injective.

(ii) If f and g are surjective, then $g \circ f : X \to Z$ is surjective.

Proof.

For part (i), assume that $f : X \to Y$ and $g : Y \to Z$ are injective. Let x_1 and x_2 be distinct elements of X. Since f is injective, $f(x_1)$ and $f(x_2)$ are distinct elements of the set Y.

Now since g is injective, $g(f(x_1))$ and $g(f(x_2))$ are distinct elements of Z. Therefore $g \circ f$ is injective as desired.

For part (ii), assume that $f : X \to Y$ and $g : Y \to Z$ are surjective. Let z be an arbitrary element of Z. Since g is surjective, there must be some element $y \in Y$ such that $g(y) = z$.

Now since f is surjective there must be an element $x \in X$ with $f(x) = y$, so $g(f(x)) = g(y) = z$ and $g \circ f : X \to Z$ is surjective as desired.

Combining the two parts of Theorem 5.14, we have the following:

Corollary 5.15.
If $f : X \to Y$ and $g : Y \to Z$ are bijective functions, then $g \circ f : X \to Z$ is bijective.

It is natural to ask whether or not the converses of the statements in Theorem 5.14 are true. We consider the converse of Theorem 5.14 *(i)* in the following example and theorem.

Example 5.16.

Define $f : \mathbb{N} \to \mathbb{R}$ by $f(n) = n^2$. Since no two natural numbers (which are all positive) have the same square, f is injective. Define $g : \mathbb{R} \to \mathbb{R}$ by $g(x) = x^2$. Since $g(-2) = 4 = g(2)$, g is not injective. Now we consider the composition $g \circ f : \mathbb{N} \to \mathbb{R}$ of these functions. For each $n \in \mathbb{N}$ we have $g(f(n)) = g(n^2) = (n^2)^2 = n^4$. Let $m, n \in \mathbb{N}$ with $m^4 = n^4$. Then

$$0 = m^4 - n^4 = (m^2 + n^2)(m - n)(m + n).$$

Because $m, n > 0$, both $m^2 + n^2$ and $m + n$ are positive. Therefore $m - n = 0$, so $m = n$. Hence $g \circ f : \mathbb{N} \to \mathbb{R}$ is injective.

This example shows that it is possible for $g \circ f$ to be injective when g is not. The next result says that if $g \circ f$ is injective, then it does follow that f is injective.

Theorem 5.17.
Let $f : X \to Y$ and $g : Y \to Z$ be functions such that $g \circ f : X \to Z$ is injective. Then $f : X \to Y$ must be injective.

Proof.

We prove that if $f : X \to Y$ is not injective, then $g \circ f : X \to Z$ is not injective, which is the contrapositive of the desired proposition. Suppose that f is not injective, then there must be elements $x_1 \neq x_2$ in X with $f(x_1) = f(x_2)$. Since g is a function, it must be true that $g(f(x_1)) = g(f(x_2))$. Since $x_1 \neq x_2$ but $(g \circ f)(x_1) = (g \circ f)(x_2)$, $g \circ f$ is not injective.

We leave the proof of the corresponding result for surjective functions as an exercise.

Theorem 5.18.
Let $f : X \to Y$ and $g : Y \to Z$ be functions such that $g \circ f : X \to Z$ is surjective. Then $g : Y \to Z$ must be surjective.

5.3.1 Inverse Relations and Inverse Functions

Definition 5.19.
If $f : X \to Y$ is a function, the *inverse relation* of f is the relation g from Y to X given by

$$g = \{(y, x) \mid (x, y) \in f\}.$$

Every function has an inverse relation. The inverse relation is not generally a function, as the following example shows. It is also different from the preimage operation $T \mapsto f^{-1}[T]$ introduced in Section 5.1.1.

Example 5.20.

Let $f : \mathbb{R} \to \mathbb{R}$ be the function defined by $f(x) = x^2$. The inverse relation of f is $g = \{(x^2, x) \mid x \in \mathbb{R}\}$. Note that g is not a function since both $(4, 2)$ and $(4, -2)$ are in g.

Definition 5.21.
If $f : X \to Y$ is a function and its inverse relation is also a function, we say that f is *invertible* and use $f^{-1} : Y \to X$ to denote the resulting *inverse function*.

Question 5.22.
When is a function $f : X \to Y$ invertible?

A real-valued function that passes the "horizontal line test" has an inverse function defined on its range. For a function $f : \mathbb{R} \to \mathbb{R}$ to have an inverse function whose domain is all of $\mathbb{R}$, it must also be surjective. One way to understand the horizontal line test is that the graph of an inverse relation is the reflection of the original graph through the line $y = x$. Since we want this reflection to pass the vertical line test, no horizontal line should meet the original graph more than once. This is the injectivity part of invertibility.

Thus the horizontal-line test captures injectivity, but invertibility also depends on the stated codomain. We begin by proving that every invertible function is injective.

Theorem 5.23.
If $f : X \to Y$ is an invertible function, then f is injective.

Proof.

Suppose that f is not injective, then there are two elements $x_1 \neq x_2$ of X such that $f(x_1) = f(x_2)$. Let $y = f(x_1)$.

By definition both (y, x_1) and (y, x_2) must be elements of the inverse relation of f. Since $x_1 \neq x_2$, this implies that the inverse relation of f is not a function. Therefore, f is not invertible.

Unfortunately, this is not a complete answer to our question because the converse of Theorem 5.23 is not true. We have found a condition that all invertible functions must satisfy, but not all functions that satisfy this condition are invertible. The following example illustrates the difficulty.

Example 5.24.

Define $f : \mathbb{N} \to \mathbb{N}$ by $f(n) = n + 1$. This function is injective (if $m + 1 = n + 1$, then $m = n$), but not invertible according to our definition. The inverse relation g contains all ordered pairs of the form $(n + 1, n)$ where $n \in \mathbb{N}$. For g to be a function from $\mathbb{N}$ to $\mathbb{N}$, every element of $\mathbb{N}$ must be the first coordinate of exactly one ordered pair in g. The problem here is that 1 is in $\mathbb{N}$, but $1 \neq n + 1$ for any $n \in \mathbb{N}$, so 1 is not the first coordinate of any of the ordered pairs in g. Therefore g is not a function from $\mathbb{N}$ to $\mathbb{N}$.

For a function $f : X \to Y$ to be invertible, every element of the codomain Y must be a first coordinate of some ordered pair in the inverse relation. That in turn means that every element of the codomain must be the second coordinate of some ordered pair in f. This is exactly what it means to say that f is surjective. This leads us to believe the following:

THEOREM 5.25.
If $f : X \to Y$ is invertible, then f is surjective.

Proof.

Assume that $f : X \to Y$ is not surjective, then there is some element $y \in Y$ so that $y \neq f(x)$ for any $x \in X$. In other words, y is not the second coordinate of an ordered pair in f. By definition then, y will not be the first coordinate of any ordered pair in the inverse relation of f. Hence the inverse relation of f is not a function from Y to X and f is not invertible.

We are now ready to give a complete answer to our question in the form of the following theorem.

THEOREM 5.26.
A function $f : X \to Y$ is invertible if and only if it is bijective.

Proof.

If f is invertible, then we may use Theorem 5.23 and Theorem 5.25 to establish the fact that f is bijective.

To see that the converse is true, suppose that $f : X \to Y$ is a bijective function. Let $g = \{(y,x) \mid (x,y) \in f\}$ be the inverse relation of f. We must show that g is a function from Y to X, i.e. that every $y \in Y$ is the first coordinate of exactly one ordered pair in g. Let $y \in Y$. Since f is surjective, $y = f(x)$ for some $x \in X$. By definition $(x,y) \in f$ implies that $(y,x) \in g$, so y is the first coordinate of at least one ordered pair in g. Now suppose that (y,x_1) and (y,x_2) are both in g. By definition this means that $f(x_1) = y$ and $f(x_2) = y$. Since f is injective this implies that $x_1 = x_2$. It follows that y cannot be the first coordinate of more than one ordered pair in g, so g is a function from Y to X and f is invertible.

Chapter 6

The Real Numbers

Introduction

We now turn our attention to developing a mathematical description of the real numbers. All of the results of calculus can be derived from these basic properties of the real number system, which is usually done in a first course in real analysis. Our basic assumptions about the real numbers are called axioms, and they are divided into several groups. Many of these axioms will be familiar to you from previous courses in algebra.

6.1 Field Axioms

Taken as a group, the field axioms tell us that the real numbers together with the operations of addition and multiplication form what is known as a *field*. Very informally, you might think of a field as something that satisfies the usual rules you encountered in high school algebra. Fields and related objects are studied in more depth in a course in abstract algebra.

Field Axioms.
The set of real numbers $\mathbb{R}$ is a collection of objects together with the two binary operations addition and multiplication that satisfy the following properties (known as the field axioms):

(i) Commutative Laws: For every $a, b \in \mathbb{R}$, $a + b = b + a$ and $ab = ba$.

(ii) Associative Laws: For every $a, b, c \in \mathbb{R}$, $(a + b) + c = a + (b + c)$ and $(ab)c = a(bc)$.

(iii) Distributive Law: For every $a, b, c \in \mathbb{R}$, $a(b + c) = ab + ac$.

(iv) Identities: There are distinct elements 0 and 1 in $\mathbb{R}$ such that $a \cdot 1 = a$ and $a + 0 = a$ for every $a \in \mathbb{R}$.

(v) Additive Inverses: For every $a \in \mathbb{R}$, there is an element $-a \in \mathbb{R}$ such that $a + (-a) = 0$.

(vi) Multiplicative Inverses: For every $a \in \mathbb{R}$ with $a \neq 0$, there is an element $a^{-1} \in \mathbb{R}$ such that $aa^{-1} = 1$.

Any set of objects satisfying all of these properties is a *field*. The rational numbers $\mathbb{Q}$ and complex numbers $\mathbb{C}$ are also fields. The additional order and completeness axioms below distinguish $\mathbb{R}$ from these examples.

A number of the familiar algebraic properties of the real numbers are immediate consequences of the field axioms. A few of them are collected in the following theorem, though there are certainly many others.

Theorem 6.1.
For any real numbers a, b, c, the following are true:

(i) If $a + b = a + c$, then $b = c$.

(ii) $-(-a) = a$.

(iii) If $a \neq 0$ and $ab = ac$, then $b = c$.

(iv) If $a \neq 0$, then $(a^{-1})^{-1} = a$.

(v) $a \cdot 0 = 0$.

(vi) $-a = (-1)a$.

(vii) If $ab = 0$, then $a = 0$ or $b = 0$.

Proof.

We prove parts *(ii)*, *(iii)*, and *(vi)*. The remaining proofs are exercises.

We first show that *(ii)* is true. To see this, note that $-(-a)$ is the additive inverse of $-a$. The reader should justify each of the following steps:

$$\begin{aligned} a + (-a) &= 0 \\ &= -a + (-(-a)) \\ &= -(-a) + (-a) \end{aligned}$$

so $a + (-a) = -(-a) + (-a)$. We apply part *(i)* to see that $a = -(-a)$ as desired.

To prove *(iii)*, suppose that $a \neq 0$ and $ab = ac$. Since $a \neq 0$, a has a multiplicative inverse a^{-1}. The reader should give a justification for each step of the following:

$$\begin{aligned} b &= 1 \cdot b \\ &= (a^{-1}a)b \\ &= a^{-1}(ab) \\ &= a^{-1}(ac) \\ &= (a^{-1}a)c \\ &= 1 \cdot c \\ &= c \end{aligned}$$

To prove *(vi)*, let a be any real number. Then:

$$\begin{aligned} a + (-1)a &= 1 \cdot a + (-1)a \\ &= a(1 + (-1)) \\ &= a \cdot 0 \\ &= 0 \end{aligned}$$

Hence $a + (-1)a = 0 = a + (-a)$ and we may apply part *(i)* to see that $(-1)a = -a$.

6.2 Order Axioms

As noted previously, there are a number of common fields. One of the differences between $\mathbb{R}$ and $\mathbb{C}$ is that we think of the real numbers as lying in order along a line. There is no natural way to organize the complex numbers in this fashion. The next group of axioms make precise what we mean when we say that the real numbers form an *ordered field*.

Order Axioms. There is an order $<$ defined on the real numbers $\mathbb{R}$ satisfying:

(i) Transitivity: If $a < b$ and $b < c$, then $a < c$.

(ii) Trichotomy: For every two real numbers a and b, exactly one of the following holds:

$$a < b \quad \text{or} \quad a = b \quad \text{or} \quad b < a$$

(iii) If $a < b$, then $a + c < b + c$ for every $c \in \mathbb{R}$.

(iv) If $a < b$ and $c > 0$, then $ac < bc$.

We derive several consequences of the fact that $\mathbb{R}$ is an ordered field.

Theorem 6.2.
The following statements are true in $\mathbb{R}$:

(i) $a > 0$ *iff* $-a < 0$.

(ii) *If* $a < b$, *then* $-a > -b$.

(iii) *If* $a \neq 0$, *then* $a^2 > 0$.

(iv) $1 > 0$.

Proof.

To prove part *(i)*, assume first that $a > 0$. We have:

$$\begin{aligned} a &> 0 \\ a + (-a) &> 0 + (-a) && \text{using Order Axiom 3} \\ 0 &> -a \end{aligned}$$

The converse is similar and left as an exercise.

We next prove part *(ii)*. The reader should determine which axiom or theorem justifies each step of the following:

$$\begin{aligned} a &< b \\ a + (-b) &< b + (-b) \\ a + (-b) &< 0 \\ ((-a) + a) + (-b) &< (-a) + 0 \\ -b &< -a \end{aligned}$$

The proofs of the remaining parts of the theorem are left to the reader.

Theorem 6.3.
Let $a, b \in \mathbb{R}$.

(i) *If* $a > 0$ *and* $b > 0$, *then* $ab > 0$.

(ii) *If* $a < 0$ *and* $b < 0$, *then* $ab > 0$.

(iii) *If* $a > 0$ *and* $b < 0$, *then* $ab < 0$.

Theorem 6.4.
Let $a \in \mathbb{R}$.

(i) *If* $a > 0$, *then* $a^{-1} > 0$.

(ii) *If* $a < 0$, *then* $a^{-1} < 0$.

Proof.

To prove part *(i)* we assume $a > 0$ and apply Trichotomy. If $a^{-1} < 0$, then $1 = aa^{-1} < 0$ by Theorem 6.3, but this contradicts Theorem 6.2, part *(iv)*. If $a^{-1} = 0$, then we have $1 = aa^{-1} = a \cdot 0 = 0$, which again contradicts Theorem 6.2, part *(iv)*. The only remaining possibility is that $a^{-1} > 0$ as desired.

The proof of *(ii)* is similar and is left for the reader.

Theorem 6.5.
Let $a \geq 0$ and $b \geq 0$. Then $a < b$ iff $a^2 < b^2$.

Proof.

Assume $a, b \geq 0$. If $a < b$, then $b - a > 0$ and $a + b > 0$, so

$$b^2 - a^2 = (b - a)(a + b) > 0.$$

Hence $a^2 < b^2$.

Conversely, suppose that $a^2 < b^2$. The possibility $a = b$ is immediately excluded. If $b < a$, the direction just proved, applied with a and b interchanged, gives $b^2 < a^2$, also a contradiction. Trichotomy therefore forces $a < b$.

6.3 Completeness of the Real Numbers

Our axioms so far ensure that $\mathbb{R}$ is an ordered field. The field $\mathbb{Q}$ of rational numbers is also an ordered field, so we need something further to distinguish between $\mathbb{R}$ and $\mathbb{Q}$. Our last axiom is based on the observation that the field $\mathbb{Q}$ has *holes*, whereas $\mathbb{R}$ does not. Let's first consider an example to clarify what we mean by this.

Example 6.6.

Let A and B be the following sets:

$$A = \{q \in \mathbb{Q} \mid q > 0 \text{ and } q^2 \leq 2\}$$

$$B = \{x \in \mathbb{R} \mid x > 0 \text{ and } x^2 \leq 2\}$$

As we will show in Theorem 6.13, there is a real number $\sqrt{2}$ whose square is 2. The number $\sqrt{2}$ is in B and $\sqrt{2}$ is larger than every other number in B.

As we showed in Theorem 2.18, there is no rational number whose square is 2. It can be shown that for every number in A, there are larger numbers that are also in A.

6.3.1 Upper and Lower Bounds

Definition 6.7. Let A be a nonempty set of real numbers. We say that a number u is an *upper bound* for A if $x \leq u$ for every $x \in A$. We say that a number m is a *lower bound* for A if $x \geq m$ for every $x \in A$. We say that A is *bounded above* if A has an upper bound, that A is *bounded below* if A has a lower bound, and that A is *bounded* if A is bounded above and below.

Example 6.8.

Let $A = \{a \in \mathbb{R} \mid a^2 \leq 2\}$, $B = [0, 7)$, $C = \mathbb{N}$, and $D = \mathbb{Z}$.

The number 2 is an upper bound for A and -2 is a lower bound for A.

The set B has an upper bound at 7 and a lower bound at 0.

The set C has a lower bound at 1, but no upper bound; the latter claim will follow from the Archimedean Property proved below.

The set D has no upper or lower bounds, as will also follow from that property.

The previous example illustrates several things. It is possible for a nonempty set to have both upper and lower bounds, just one bound, or no bounds at all. Upper and lower bounds may or may not be elements of the set. Finally, upper and lower bounds are not unique. Transitivity implies that if M is an upper bound for a set A, then every number larger than M is also an upper bound for A. Similarly, if m is a lower bound for A then every number smaller than m is a lower bound for A.

Consider the set $A = \{a \in \mathbb{R} \mid a^2 \leq 2\}$ from the previous example again. We claimed that 2 is an upper bound, and that is certainly true, but note that $3/2$ is also an upper bound. In some sense this smaller upper bound is a "better" bound for A because it puts a tighter restriction on the size of the elements of A. This is approximately like saying that Grand Forks is a better way to describe the location of UND than North Dakota. In this sense the best possible upper bound would be the smallest one, if there is one. In this case, A has a smallest upper bound in $\mathbb{R}$, namely $\sqrt{2}$. The analogous rational set $\{q \in \mathbb{Q} \mid q^2 \leq 2\}$ has no least upper bound in $\mathbb{Q}$. This is the basic difference between $\mathbb{Q}$ and $\mathbb{R}$ that we wish to capture in an axiom. First, we need a couple more definitions.

Definition 6.9. Let A be a nonempty subset of $\mathbb{R}$. We say that a number u is a *least upper bound* (LUB) for A if both of the following are true:

(i) For every $a \in A$, $a \leq u$.

(ii) For every upper bound b of A, $u \leq b$.

We say that a number m is a *greatest lower bound* (GLB) for A if both of the following are true:

(i) For every $a \in A$, $a \geq m$.

(ii) If b is a lower bound for A, then $m \geq b$.

Note that a LUB is sometimes called a *supremum* and a GLB is sometimes called an *infimum*.

It is not hard to prove that, unlike upper bounds, a set can have only one least upper bound. This fact is made explicit in the following theorem, whose proof is left as a homework exercise.

Theorem 6.10.
Let A be a nonempty subset of $\mathbb{R}$. If u and v are least upper bounds for A, then $u = v$.

The following theorem says that the least upper bound of a set must in some sense be "close to" the set.

Theorem 6.11.
Let A be a nonempty subset of $\mathbb{R}$ and let u be the least upper bound for A. If $x < u$, then there is an element $a \in A$ such that $x < a$.

Proof.

Suppose that u is the least upper bound for A and that $x < u$. Assume that there is no $a \in A$ such that $x < a$, then $a \leq x$ for every $a \in A$ by Trichotomy. It follows by definition that x is an upper bound for A, but this contradicts the fact that u is the least upper bound for A since $x < u$.

We are now ready to state our final axiom, which says that $\mathbb{R}$ is *complete*.

The Completeness Axiom. Let A be a nonempty subset of $\mathbb{R}$. If A has an upper bound, then A has a least upper bound.

There are a number of consequences of the Completeness Axiom, most of which are beyond the scope of this text. We will look at only a couple of them. We begin with the fairly intuitive-seeming fact that for any real number x, there is a natural number larger than x.

Theorem 6.12.
(Archimedean Property) *If $x \in \mathbb{R}$, then there is an $n \in \mathbb{N}$ such that $n > x$.*

Proof.

Suppose to the contrary that some $x \in \mathbb{R}$ satisfies $n \leq x$ for every $n \in \mathbb{N}$. Then $\mathbb{N}$ is bounded above and, by completeness, has a least upper bound m. Since $m - 1 < m$, Theorem 6.11 gives an $n \in \mathbb{N}$ with $m-1 < n$. Thus $m < n+1$, while $n+1 \in \mathbb{N}$, contradicting that m is an upper bound.

We previously commented, without proof, that the Completeness Axiom would allow us to distinguish $\mathbb{R}$ from $\mathbb{Q}$. We will now make this explicit by proving that there is at least one real number that is not rational.[1] By Theorem 2.18, there is no rational number whose square is 2. We now show that the Completeness Axiom implies that there must be a real number x with $x^2 = 2$.

Theorem 6.13.
There is a number $x \in \mathbb{R}$ such that $x^2 = 2$.

Proof.

Let

$$A = \{a \in \mathbb{R} \mid a \geq 0 \text{ and } a^2 \leq 2\}.$$

[1]In fact there are more irrational numbers than there are rational numbers, as we shall see in the final chapter.

The set A is nonempty because $1 \in A$. It is bounded above by 2: if $a > 2$, then $a^2 > 4 > 2$, so $a \notin A$. Completeness therefore gives a least upper bound $x = \sup A$. Because $1 \in A$, we have $x \geq 1$. We use trichotomy to prove that $x^2 = 2$.

First suppose that $x^2 < 2$. The number $(2x+1)/(2-x^2)$ is positive. By the Archimedean Property, choose $n \in \mathbb{N}$ such that

$$n > \max\left\{1, \frac{2x+1}{2-x^2}\right\}.$$

Then $1/n < 1$ and $1/n < (2-x^2)/(2x+1)$. Consequently

$$\begin{aligned}
\left(x + \frac{1}{n}\right)^2 &= x^2 + \frac{2x}{n} + \frac{1}{n^2} \\
&= x^2 + \frac{1}{n}\left(2x + \frac{1}{n}\right) \\
&< x^2 + \frac{1}{n}(2x+1) \\
&< x^2 + (2 - x^2) \\
&= 2
\end{aligned}$$

Thus $x + 1/n \in A$, contradicting that x is an upper bound. Hence $x^2 < 2$ is impossible.

Next suppose that $x^2 > 2$. Choose $m \in \mathbb{N}$ with

$$m > \max\left\{1, \frac{2x}{x^2-2}\right\}.$$

Because $x \geq 1$ and $1/m < 1$, the number $y = x - 1/m$ is nonnegative. Also

$$\begin{aligned} y^2 = \left(x - \frac{1}{m}\right)^2 &= x^2 - \frac{2x}{m} + \frac{1}{m^2} \\ &> x^2 - \frac{2x}{m} \\ &> x^2 - (2x)\frac{x^2 - 2}{2x} \\ &= 2 \end{aligned}$$

We claim that y is an upper bound for A. If some $a \in A$ satisfied $a > y$, then $a, y \geq 0$ and Theorem 6.5 would give $a^2 > y^2 > 2$, contrary to $a \in A$. Therefore every element of A is at most y. But $y < x$, contradicting that x is the least upper bound. Hence $x^2 > 2$ is impossible.

Trichotomy leaves only $x^2 = 2$.

Chapter 7

Introduction to Cardinality

Introduction

What do we mean when we say that there are *four* suits in a standard deck of cards? More generally, what does it mean to count any collection of objects? Since you may no longer actually think about counting, it may help to watch a child who is just learning to count. Given four objects to count, the child is likely to point to each object in turn and count "one, two, three, four." In other words, the child is explicitly constructing a bijective function between the objects she is counting and the elements of the set $\{1, 2, 3, 4\}$. Our goal in this chapter is to extend this idea to infinite sets.

7.1 The Cardinality of a Set

For sets A and B, write $A \equiv B$ if there is a bijective function $f : A \to B$. In this case we say that A and B are *equinumerous* or that they have the same *cardinality*. Write $A \preceq B$ if there is an injective function $f : A \to B$; equivalently, $B \succeq A$. We write $A \prec B$ to indicate that $A \preceq B$ and $A \not\equiv B$. It is also common to write $|A| = |B|$ rather than $A \equiv B$.

Theorem 7.1.
For any set $\mathcal{C}$ whose elements are sets, equinumerosity restricts to an equivalence relation on $\mathcal{C}$.

Proof.

Let A, B, and C be arbitrary elements of $\mathcal{C}$. Since the identity function on any set is a bijection, $A \equiv A$ and $\equiv$ is reflexive.

If $A \equiv B$, then there is a bijection $f : A \to B$. By Theorem 5.26, its inverse $f^{-1} : B \to A$ is a function; it is bijective because an inverse interchanges the roles of inputs and outputs. Hence $B \equiv A$ and $\equiv$ is symmetric.

To see that $\equiv$ is transitive, suppose that $A \equiv B$ and $B \equiv C$. By definition there are bijective functions $f : A \to B$ and $g : B \to C$. Now apply Corollary 5.15 to see that $g \circ f : A \to C$ is a bijective function. Hence $A \equiv C$ and $\equiv$ is transitive. Therefore, $\equiv$ is an equivalence relation as desired.

Note that Theorem 7.1 allows us to say that two sets A and B have the same cardinality if we are able to find a bijection from A to B or a bijection from B to A.

Theorem 7.2.
On any set $\mathcal{C}$ whose elements are sets, the relation $\preceq$ is reflexive and transitive.

Proof.

Let A, B, and C be elements of $\mathcal{C}$. Since the identity function on any set is injective, $A \preceq A$ and $\preceq$ is reflexive. To see that $\preceq$ is transitive, suppose that $A \preceq B$ and $B \preceq C$. By definition there are injections $f : A \to B$ and $g : B \to C$. We apply Theorem 5.14 to see that the composition $g \circ f : A \to C$ is also injective, hence $A \preceq C$.

While we would not expect $\preceq$ to be symmetric, we can ask for a cardinal analogue of antisymmetry. Mutual injections do not force two sets to be literally equal, but do they force the sets to be equinumerous? Georg Cantor (1845–1918) was interested in this question, and Felix Bernstein (1878–1956) proved that the answer is yes. The result is usually known as the Cantor–Bernstein Theorem.

Theorem 7.3 (Cantor–Bernstein).
If $A \preceq B$ *and* $B \preceq A$, *then* $A \equiv B$.

We prove this theorem in Chapter A. The following example uses the same kind of chain-shifting idea to construct a bijection between the open interval $(-1, 1)$ and the closed interval $[-1, 1]$.

Example 7.4.

Define the functions $f : (-1, 1) \to [-1, 1]$ and $g : [-1, 1] \to (-1, 1)$ by $f(x) = x$ and $g(x) = x/2$, respectively. It is easy to show that both of these functions are injective, so $(-1, 1) \preceq [-1, 1]$ and $[-1, 1] \preceq (-1, 1)$. The Cantor–Bernstein Theorem implies that there must be a bijection between the open interval $(-1, 1)$ and the closed interval $[-1, 1]$, but the theorem itself doesn't tell us how to find such a bijection. We construct such a bijection here.

We wish to find a bijective function $h : (-1, 1) \to [-1, 1]$. For most elements $x \in (-1, 1)$ we want $h(x) = f(x) = x$. Unfortunately, if we let $h(x) = f(x)$ for all x, then the function is not bijective because $-1 \neq f(x)$ and $1 \neq f(x)$ for all $x \in (-1, 1)$. We use the function g to help us fix this problem. Consider first the element $1 \in [-1, 1]$. While $1 \neq f(x)$ for any x, there is an element of the open interval associated with 1 by g. In particular $g(1) = 1/2$. We could define h so that $h(1/2) = 1$ and $h(x) = f(x)$ for other $x \in (-1, 1)$, but that creates a new problem. Now 1 is in the image of the function, but $1/2$ is not. To fix this we let $h(1/4) = 1/2$, because $g(1/2) = 1/4$. Of course, now $1/4$ is not in the image of h. We continue fixing one problem at a time using g, each time creating a new problem. Naturally, we will need to do the same with -1.

These particular functions are simple enough that we can write down a formula for the function h that we end up with. We end up defining $h : (-1, 1) \to [-1, 1]$ by

$$h(x) = \begin{cases} 2^{-(n-1)} & \text{if } x = 2^{-n} \text{ for some } n \in \mathbb{N} \\ -2^{-(n-1)} & \text{if } x = -2^{-n} \text{ for some } n \in \mathbb{N} \\ x & \text{otherwise} \end{cases}$$

We claim that this function is the desired bijection.

We first show that h is surjective. To see this, let $t \in [-1, 1]$. If $t = 2^{-(n-1)}$ for some $n \in \mathbb{N}$, then $h(2^{-n}) = t$. If $t = -2^{-(n-1)}$ for some $n \in \mathbb{N}$, then $h(-2^{-n}) = t$. For any other $t \in [-1, 1]$ we have $h(t) = t$. In any case $t = h(x)$ for some $x \in (-1, 1)$, so h is surjective.

To see that h is injective, suppose that $x \neq y$ are both elements of $(-1, 1)$. We consider several cases.

Case 1. $x = 2^{-n}$, $y = 2^{-k}$ for some $n, k \in \mathbb{N}$. Since $x \neq y$, it follows that $n \neq k$. Hence $n - 1 \neq k - 1$ and we have $h(x) = 2^{-(n-1)} \neq 2^{-(k-1)} = h(y)$.

Case 2. $x = -2^{-n}$, $y = -2^{-k}$ for some $n, k \in \mathbb{N}$. Since $x \neq y$, it follows that $n \neq k$. Hence $n - 1 \neq k - 1$ and we have $h(x) = -2^{-(n-1)} \neq -2^{-(k-1)} = h(y)$.

Case 3. $x = 2^{-n}$, $y = -2^{-k}$ for some $n, k \in \mathbb{N}$. In this case we have $h(x) = 2^{-(n-1)} \neq -2^{-(k-1)} = h(y)$.

Case 4. $x = -2^{-n}$, $y = 2^{-k}$ for some $n, k \in \mathbb{N}$. In this case we have $h(x) = -2^{-(n-1)} \neq 2^{-(k-1)} = h(y)$.

Case 5. $x = \pm 2^{-n}$ for some $n \in \mathbb{N}$ and $y \neq \pm 2^{-k}$ for any $k \in \mathbb{N}$. In this case we have $h(x) = \pm 2^{-(n-1)} \neq y = h(y)$.

Case 6. $y = \pm 2^{-n}$ for some $n \in \mathbb{N}$ and $x \neq \pm 2^{-k}$ for any $k \in \mathbb{N}$. In this case we have $h(x) = x \neq \pm 2^{-(n-1)} = h(y)$.

Case 7. $x \neq \pm 2^{-n}$ and $y \neq \pm 2^{-n}$ for any $n \in \mathbb{N}$. In this case we have $h(x) = x \neq y = h(y)$.

In all cases we have $h(x) \neq h(y)$, so h is injective.

7.2 Finite Sets

Question.
What does it mean to say that a set A is finite?

At first glance, this may seem like something you've known for a long time. Don't we just mean that we can count the elements of A? If so, is the number of cells in your body finite? Can you count them? Maybe the answer to our question isn't quite so obvious after all.

For each $n \in \mathbb{N} \cup \{0\}$, define

$$\mathrm{I}_0 = \varnothing \qquad \text{and} \qquad \mathrm{I}_n = \{1, 2, \ldots, n\} \quad \text{when } n \in \mathbb{N}.$$

Thus $\mathrm{I}_4 = \{1, 2, 3, 4\}$. These sets let us formalize the idea of counting while handling the empty set without a separate convention.

Definition 7.5.
Let A be a set and let $n \in \mathbb{N} \cup \{0\}$. We say that A has n elements if $\mathrm{A} \equiv \mathrm{I}_n$. The set A is *finite* if $\mathrm{A} \equiv \mathrm{I}_n$ for some $n \in \mathbb{N} \cup \{0\}$, and it is *infinite* if it is not finite.

Theorem 7.6. *The set $\mathbb{N}$ is infinite.*

Proof.

If $\mathbb{N}$ were finite, it could be listed as $\{a_1, a_2, \ldots, a_n\}$ for some $n \in \mathbb{N}$; the empty case is impossible. Every nonempty finite list of natural numbers has a largest entry, as follows by induction on its length. If M were the largest member of this list, then $\mathrm{M} + 1 \in \mathbb{N}$ would be absent from the list, a contradiction.

Composing a bijection $\mathrm{A} \to \mathrm{I}_n$ with the inclusion $\mathrm{I}_n \hookrightarrow \mathbb{N}$ gives the following.

Theorem 7.7.
If A is finite, then $\mathrm{A} \preceq \mathbb{N}$.

Now that we have a definition of finite, let's reconsider the set C of cells in your body. Is C finite? If so, for which n is $\mathrm{C} \equiv \mathrm{I}_n$? We still can't really count them, so perhaps we've just obscured the question rather than answering it. Scientists estimate that there are about 10,000,000,000,000 cells in the average adult human body. That's not an actual count of the number of cells in any individual human body, though. These kinds of estimates are based on the sizes of various kinds of cells and the approximate proportion of each kind of cell in the body. In fact, we could determine the maximum number of cells that might be in a person's body by figuring out how many of the smallest kinds of cells would be required to build a body of a particular volume, or weight, etc. It seems reasonable to think that we could say a set was finite if we were sure it had at most n elements for some natural number n. That is the intent of the next result.

Theorem 7.8.
If $A \preceq I_n$ for some $n \in \mathbb{N} \cup \{0\}$, then A is finite.

Suppose there is an injection $f : A \to I_n$. Our definition of finite asks for a bijection with some I_k, while f need not be surjective. The next lemma removes one unused point from the codomain at a time.

Lemma 7.9.
Let $n \in \mathbb{N}$. If $f : A \to I_n$ is injective but not surjective, then there is an injection $g : A \to I_{n-1}$.

Proof.

Since f is not surjective, $B = I_n \setminus f[A]$ is nonempty. Choose $b \in B$. If $b = n$, then the range of f is already in I_{n-1}, so set $g = f$. If $b \neq n$, define

$$g(a) = \begin{cases} f(a), & f(a) \neq n, \\ b, & f(a) = n. \end{cases}$$

In either case g maps A into I_{n-1}. It remains to be shown that g is injective. Suppose that $a_1, a_2 \in A$ and that $g(a_1) = g(a_2) = m$. If $m = b$, then $b \notin f[A]$, so the first branch in the definition of g cannot produce b; consequently $f(a_1) = n = f(a_2)$. If $m \neq b$, then the definition of g implies that $f(a_1) = m = f(a_2)$. In either case $f(a_1) = f(a_2)$, so $a_1 = a_2$ because f is injective. Therefore g is injective.

We will now prove the theorem.

Proof of Theorem 7.8.

If $A = \varnothing$, then $A \equiv I_0$ and the result is immediate. Assume $A \neq \varnothing$. The hypothesis then forces $n \geq 1$. Let $f_0 : A \to I_n$ be injective. If f_0 is surjective, it is the required bijection. Otherwise, Lemma 7.9 gives an injection $f_1 : A \to I_{n-1}$.

If f_1 is surjective, it is a bijection. If not, apply the lemma again to obtain $f_2 : A \to I_{n-2}$.

Continue. If one of the injections $f_i : A \to I_{n-i}$ is surjective, it is a bijection. The process must stop by $i = n - 1$: an injection from the nonempty set A into $I_1 = \{1\}$ is automatically surjective.

Thus $A \equiv I_{n-i}$ for some $0 \leq i \leq n - 1$, so A is finite.

Consequently, if $\mathbb{N} \preceq A$, then A is infinite.

If A were finite, a bijection from A to some I_n, composed with an injection from $\mathbb{N}$ to A, would give an injection $\mathbb{N} \to I_n$. By Theorem 7.8, this would make $\mathbb{N}$ finite, contradicting Theorem 7.6.

We conclude this section with a question, the answer to which may seem obvious to you.

Question 7.10.
If $m, n \in \mathbb{N} \cup \{0\}$ and $m \neq n$, can you prove that $\mathrm{I}_m \not\approx \mathrm{I}_n$?

We prove by induction on m the stronger fact that I_k cannot inject into I_m whenever $k > m$. For $m = 0$, the codomain is empty while the domain is nonempty, so the claim is immediate.

Assume the claim holds for m, let $k > m+1$, and suppose that $f : \mathrm{I}_k \to \mathrm{I}_{m+1}$ is injective. If $m+1$ is not in the range of f, then f is an injection into I_m, contrary to the inductive hypothesis. If $m+1$ is in the range, remove its unique preimage from I_k and restrict f. After relabeling the remaining $k-1$ elements as I_{k-1}, this gives an injection $\mathrm{I}_{k-1} \to \mathrm{I}_m$. Since $k-1 > m$, this again contradicts the inductive hypothesis.

Thus no injection $\mathrm{I}_n \to \mathrm{I}_m$ exists when $m < n$. A bijection $\mathrm{I}_m \to \mathrm{I}_n$ would have an inverse of exactly that kind. If $n < m$, interchange the names of m and n.

7.3 Denumerable Sets

Georg Cantor was able to define a complete system of infinite numbers and of arithmetic on those numbers. We will not discuss his system here, but we will look at one particular kind of infinite number that is important in many areas of mathematics.

Theorem 7.11.
If A *is an infinite set and* B *is a finite subset of* A, *then the set* $A \setminus B$ *is infinite.*

Proof.

Suppose to the contrary that $A \setminus B$ is finite. Choose bijections $f : B \to I_m$ and $g : A \setminus B \to I_n$ for suitable $m, n \in \mathbb{N} \cup \{0\}$. Define $h : A \to I_{m+n}$ by

$$h(a) = \begin{cases} f(a), & a \in B, \\ m + g(a), & a \in A \setminus B. \end{cases}$$

The two branches have disjoint ranges, and each is injective, so h is injective. Theorem 7.8 would then make A finite, contrary to the hypothesis. Thus $A \setminus B$ is infinite.

Definition 7.12.
Let A be a set. We say that A is:

- *denumerable* if $A \equiv \mathbb{N}$.
- *countable* if A is either finite or denumerable.
- *uncountable* if A is infinite and not denumerable.

Example 7.13.

The set $A = \{2k \mid k \in \mathbb{N}\}$ of even natural numbers is denumerable. To see this, define the function $f : \mathbb{N} \to A$ by $f(n) = 2n$. It is routine to check that f is a bijection, so $\mathbb{N} \equiv A$ as desired.

The preceding example points out an important difference between finite and infinite sets. The set A is a proper subset of $\mathbb{N}$, but has the same cardinality as $\mathbb{N}$. A set equinumerous with one of its proper subsets is called *Dedekind-infinite*. To see why a finite set cannot have this property, suppose $A \equiv I_n$ and $B \subsetneq A$. The image of B under a bijection $A \to I_n$ is a proper subset S of I_n; in particular, $n \geq 1$. The inclusion $S \to I_n$ is not surjective, so Lemma 7.9 gives an injection $S \to I_{n-1}$. If $B \equiv A$, this would produce an injection $I_n \to I_{n-1}$, contradicting the induction in the completion of Question 7.10. Therefore every Dedekind-infinite set is infinite. The converse holds in standard ZFC set theory, but its proof uses a form of the Axiom of Choice; without Choice, the two notions need not coincide.

We would like to determine which of our results about finite sets are also true for denumerable sets. If A and B are denumerable, must $A \cup B$ also be denumerable? Are subsets of denumerable sets denumerable? Is there an analog of Theorem 7.8? Are there other ways to tell that a set is denumerable?

Theorem 7.14.
If A is a denumerable set and $B \equiv A$, then B is denumerable.

Proof. By definition we have $A \equiv \mathbb{N}$. Since $\equiv$ is an equivalence relation, it follows that $B \equiv \mathbb{N}$ as desired.

Theorem 7.15.
If $A \subseteq \mathbb{N}$, *then* A *is countable.*

Proof.

If A is finite, then A is countable by definition. Assume that A is infinite. We define a function $f : \mathbb{N} \to A$ as follows. Recall that every nonempty subset of $\mathbb{N}$ has a smallest element. Let $f(1)$ be the smallest element of $A_0 = A$. Since A is infinite, the set $A_1 = A \setminus \{f(1)\}$ is nonempty, so we may define $f(2)$ to be the smallest element of A_1. Continuing recursively, suppose that we have defined $f(1), \ldots, f(n)$ for some $n \in \mathbb{N}$. Let $A_n = A \setminus \{f(1), \ldots, f(n)\}$. Since $\{f(1), \ldots, f(n)\}$ is finite, A_n is infinite by Theorem 7.11. In particular, A_n is nonempty and we may define $f(n+1)$ to be the smallest element of this set. Before showing that f is bijective we note the following facts that follow immediately from our construction:

1. For every natural number n, $A \setminus A_n = \{f(1), \ldots, f(n)\}$.
2. For every natural number n, $f(n) \geq n$.
3. For all natural numbers m, n, if $f(n) \in A_m$ then $m < n$.
4. For all natural numbers $m < n$, $f(n) \in A_m$.
5. For all natural numbers $m \leq n$, $f(m) \notin A_n$.

For any two natural numbers $m < n$ we have shown that $f(n) \in A_m$ and $f(m) \notin A_m$, so $f(n) \neq f(m)$ and f is injective. To see that f is surjective, let $k \in A$. We must show that $k = f(m)$ for some $m \in \mathbb{N}$. If $k = f(k)$, we are done. Otherwise we have $f(k) > k$. Now $f(k)$ is the smallest element of A_{k-1} and $k < f(k)$, so $k \notin A_{k-1}$. We also know that $k \in A$, so it follows that $k \in A \setminus A_{k-1} = \{f(1), \ldots, f(k-1)\}$. Therefore $k = f(n)$ for some $1 \leq n < k$ and f is surjective as desired.

Corollary 7.16. *A set* A *is countable if and only if* $A \preceq \mathbb{N}$.

Proof.

Suppose first that A is countable. If A is finite, then $A \preceq \mathbb{N}$ by Theorem 7.7. If A is denumerable, a bijection from A to $\mathbb{N}$ is in particular an injection.

Conversely, suppose that $f : A \to \mathbb{N}$ is injective. After restricting its codomain, $f : A \to f[A]$ is a bijection, and Theorem 7.15 shows that $f[A]$ is countable. If $f[A]$ is finite, then A is finite; if $f[A]$ is denumerable, Theorem 7.14 shows that A is denumerable. Thus A is countable.

Corollary 7.17. *A subset of a countable set is countable.*

Proof.

Let $C \subseteq A$, where A is countable. The inclusion $C \to A$ is injective, and Corollary 7.16 supplies an injection $A \to \mathbb{N}$. Their composition is an injection $C \to \mathbb{N}$, so Corollary 7.16 shows that C is countable.

Thus a set A is countable exactly when there is an injection from A to $\mathbb{N}$. If A is nonempty, this is also equivalent to finding a surjection from $\mathbb{N}$ onto A. In one direction, invert an injection on its range and send unused natural numbers to a fixed element of A; in the other, assign each $a \in A$ the least natural number mapping to it. The nonempty hypothesis is necessary: there is no function from $\mathbb{N}$ onto $\varnothing$.

Corollary 7.18.
A set A is countable if either of the following is true:

(i) *There is an injection* $f : A \to \mathbb{N}$.

(ii) $A \neq \varnothing$ *and there is a surjection* $f : \mathbb{N} \to A$.

Theorem 7.19.
The union of two denumerable sets is denumerable.

Proof.

Suppose that we are given any two denumerable sets A and B. By definition there are bijective functions $f : A \to \mathbb{N}$ and $g : B \to \mathbb{N}$. We define a function $h : A \cup B \to \mathbb{N}$ by the following rule:

$$h(x) = \begin{cases} 2f(x) & \text{if } x \in A \\ 2g(x) + 1 & \text{if } x \notin A \end{cases}$$

We claim that h is an injection. To see this, let $x \neq y$ be two elements of $A \cup B$. If $x \in A$ and $y \notin A$, then $h(x) \neq h(y)$ since $h(x)$ is even and $h(y)$ is odd. Similarly if $x \notin A$ and $y \in A$, then $h(x) \neq h(y)$. If x and y are both in A, then $h(x) = 2f(x)$ and $h(y) = 2f(y)$, so $h(x) \neq h(y)$ because f is injective. Finally, if neither x nor y are in A, then $h(x) = 2g(x) + 1$ and $h(y) = 2g(y) + 1$, so $h(x) \neq h(y)$ because g is injective. In any case, we have shown that $h(x) \neq h(y)$ so $h : A \cup B \to \mathbb{N}$ is injective.

We have shown that $A \cup B \preceq \mathbb{N}$, so $A \cup B$ is countable. It is actually denumerable: the inclusion $A \hookrightarrow A \cup B$ is injective, and $\mathbb{N} \preceq A$ because A is denumerable. Transitivity gives $\mathbb{N} \preceq A \cup B$. Cantor–Bernstein now yields $A \cup B \equiv \mathbb{N}$.

Using Mathematical Induction on the number of sets, we obtain:

Corollary 7.20.
The union of any positive finite number of denumerable sets is denumerable.

Corollary 7.21. *The union of two countable sets is countable.*

Proof.

Let A and B be countable. By Corollary 7.16, choose injections $f : A \to \mathbb{N}$ and $g : B \to \mathbb{N}$. Define

$$h(x) = \begin{cases} 2f(x), & x \in A, \\ 2g(x) + 1, & x \in B \setminus A. \end{cases}$$

The two branches have disjoint ranges and each branch is injective, so $h : A \cup B \to \mathbb{N}$ is injective. Corollary 7.16 now shows that $A \cup B$ is countable.

Theorem 7.22.
The set $\mathbb{N} \times \mathbb{N}$ is denumerable.

Proof.

The map $n \mapsto (n, 1)$ injects $\mathbb{N}$ into $\mathbb{N} \times \mathbb{N}$, so the consequence following Theorem 7.8 shows that the product is infinite. It therefore suffices to prove that it is countable. Define the function $f : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ by $f(a, b) = 2^{a-1}(2b - 1)$. We will show that f is injective, then apply Corollary 7.18. To do so, suppose that

$$2^{a-1}(2b-1) = 2^{c-1}(2d-1) \tag{7.1}$$

for $(a, b), (c, d) \in \mathbb{N} \times \mathbb{N}$. We show first that $a = c$. If not, then we may assume without loss of generality that $a < c$. Dividing both sides of Equation (7.1) by 2^{a-1} yields $(2b - 1) = 2^{c-a}(2d - 1)$. But this is a contradiction since the quantity on the left side of the equation is odd and the quantity on the right is even. Since $a = c$, we may reduce Equation (7.1) to $(2b - 1) = (2d - 1)$, from which it follows that $b = d$. Now $(a, b) = (c, d)$ and f is injective as desired.[1]

[1]While it is not necessary for the purposes of this example, it is not too difficult to show that the function f defined here is actually bijective.

7.3.1 The Rational Numbers

At first glance it may seem that there are more rational numbers than there are natural numbers. After all, there are infinitely many rational numbers between any two natural numbers. One of Cantor's accomplishments was to show that the set of rational numbers is actually denumerable. Our intuition developed from years of working with finite sets just doesn't serve us very well when working with infinite sets.

Lemma 7.23.
The set $\mathbb{Q}^+$ of positive rational numbers is countable.

Proof.

Let $F = \mathbb{N} \times \mathbb{N}$. We may write a pair $(a, b) \in F$ as the formal fraction a/b, but distinct pairs remain distinct even when they represent the same rational number. By Theorem 7.22, F is denumerable. Define

$$p : F \to \mathbb{Q}^+, \qquad p(a, b) = \frac{a}{b},$$

where the expression on the right is now interpreted as a rational number. Every positive rational number has such a representation, so p is surjective. Choose a bijection $e : \mathbb{N} \to F$. Then $p \circ e : \mathbb{N} \to \mathbb{Q}^+$ is surjective, and Corollary 7.18 shows that $\mathbb{Q}^+$ is countable.

Since the function taking every positive rational to its additive inverse is bijective, the following corollary is immediate from the lemma.

Corollary 7.24.
The set $\mathbb{Q}^-$ of negative rational numbers is countable.

We know that $\mathbb{Q}^+$, $\mathbb{Q}^-$, and $\{0\}$ are all countable sets. Applying Corollary 7.21 twice, we conclude:

Theorem 7.25.
The set $\mathbb{Q}$ of rational numbers is denumerable.

Proof.

The preceding results show that $\mathbb{Q}$ is countable. The inclusion $\mathbb{N} \hookrightarrow \mathbb{Q}$, together with the consequence following Theorem 7.8, shows that $\mathbb{Q}$ is infinite. Therefore $\mathbb{Q}$ is denumerable.

7.3.2 The Real Numbers

By this point it may seem possible that all sets are countable, making denumerability a useless distinction. We will show that this is not true by proving that the real numbers are uncountable. A real number can have two decimal expansions; for example, $1.000\ldots = 0.999\ldots$. To make decimal digits unambiguous, we choose for each real number its *canonical* expansion: the one that is not eventually all 9s.

Theorem 7.26.
The set $\mathbb{R}$ of real numbers is uncountable.

Proof.

Suppose to the contrary that $\mathbb{R}$ is countable. Since $\mathbb{R}$ is nonempty, there would be a surjection $f : \mathbb{N} \to \mathbb{R}$. Let $f(m)_n$ denote the nth digit after the decimal point in the canonical expansion of $f(m)$. Define digits

$$y_n = \begin{cases} 2, & \text{if } f(n)_n = 1, \\ 1, & \text{if } f(n)_n \neq 1, \end{cases}$$

and let $y = 0.y_1y_2y_3\ldots$. Because every y_n is 1 or 2, this expansion is canonical and represents a number $y \in (0,1)$. For every $n \in \mathbb{N}$, the nth digit of y differs from the nth digit of $f(n)$. Thus $y \neq f(n)$ for every n, contradicting surjectivity.

Appendix A

The Cantor–Bernstein Theorem

The relation $X \preceq Y$ means that X injects into Y. It is reflexive and transitive, but it is not literally antisymmetric on sets: distinct sets can have the same cardinality. The Cantor–Bernstein Theorem supplies the appropriate cardinal version of antisymmetry:

$$X \preceq Y \text{ and } Y \preceq X \quad \Longrightarrow \quad X \equiv Y.$$

The proof below is constructive once the two injections are given.

A.1 The Forward-Chain Construction

Assume that $f : X \to Y$ and $g : Y \to X$ are injective. Some points of X have no predecessor under g; begin a forward chain at each such point. Define

$$A_0 = X \setminus g[Y], \qquad A_{n+1} = g\big[f[A_n]\big] \quad (n = 0, 1, 2, \ldots),$$

and let

$$A = \bigcup_{n=0}^{\infty} A_n = \{x \in X \mid x \in A_n \text{ for some } n \in \mathbb{N} \cup \{0\}\}.$$

On the chains in A we use f in the forward direction. Off those chains, every point has a unique predecessor under g, so we use that predecessor.

Proof of Theorem 7.3.

For $x \in X \setminus A$, we have $x \notin A_0$, hence $x \in g[Y]$. Thus there is a $y \in Y$ with $g(y) = x$, and this y is unique because g is injective. We may therefore define $h : X \to Y$ by

$$h(x) = \begin{cases} f(x), & x \in A, \\ \text{the unique } y \in Y \text{ such that } g(y) = x, & x \notin A. \end{cases}$$

We first prove that h is injective. Suppose $h(x_1) = h(x_2)$.

If $x_1, x_2 \in A$, then $f(x_1) = f(x_2)$, so $x_1 = x_2$ because f is injective.

If $x_1, x_2 \notin A$, then

$$x_1 = g(h(x_1)) = g(h(x_2)) = x_2.$$

It remains to rule out a mixed case. Suppose $x_1 \in A$ and $x_2 \notin A$. Choose $n \geq 0$ such that $x_1 \in A_n$. If $h(x_1) = h(x_2)$, then

$$x_2 = g(h(x_2)) = g(h(x_1)) = g(f(x_1)).$$

Hence $x_2 \in g[f[A_n]] = A_{n+1} \subseteq A$, a contradiction. The other mixed case is symmetric. Therefore h is injective.

To prove surjectivity, let $y \in Y$. If $g(y) \notin A$, then the second clause in the definition gives

$$h(g(y)) = y.$$

Now suppose $g(y) \in A$. Since $g(y) \in g[Y]$, it cannot lie in $A_0 = X \setminus g[Y]$. Therefore $g(y) \in A_{n+1}$ for some $n \geq 0$. By the definition of A_{n+1}, there is an $x \in A_n$ such that

$$g(y) = g(f(x)).$$

Injectivity of g gives $y = f(x)$, and because $x \in A$, $h(x) = f(x) = y$. Thus every $y \in Y$ has a preimage under h, so h is surjective. The function h is both injective and surjective; hence it is the desired bijection $X \to Y$.

A.2 Why the Construction Works

The set A_0 records where the backward search through g must stop. Each A_{n+1} advances those starting points one step through f and then g. Using f on every such forward chain prevents a missing predecessor from causing a gap. Everywhere else, the g-chains extend backward indefinitely, so reversing g is safe. The argument also covers empty sets without a separate case.